Project Name

Project ID

Project Proposal Report

Nethmi I W D A – IT22082756

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

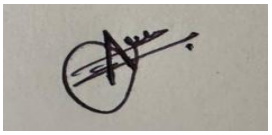
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

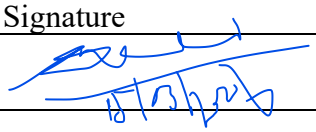
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Nethmi I W D A	**IT22082756**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
SMB Harshanath		

ABSTRACT

Road infrastructure plays a vital role in transportation safety, economic activities, and daily mobility. However, road inspection in many developing countries, including Sri Lanka, is mainly conducted through manual surveys by road officers. This process is time-consuming, inconsistent, and highly dependent on human judgment. As a result, early-stage road defects such as potholes, cracks, surface erosion, and uneven surfaces are often not detected in time, leading to increased repair costs, vehicle damage, and road accidents.

Although several modern studies have introduced computer vision techniques for detecting road defects, most existing systems focus only on defect detection and do not provide a complete maintenance decision-support solution. They rarely integrate repair area estimation, material calculation, and repair time prediction within a single system.

This research proposes **ROAD INSPECTOR**, an intelligent road inspection and repair estimation system that uses image processing and sensor data to detect road defects and support maintenance planning. The system captures road images using camera modules and applies computer vision algorithms to identify defects such as potholes, cracks, and erosion. It further estimates the affected repair area, calculates required materials, and predicts repair duration by analyzing environmental and traffic conditions.

The proposed system will utilize technologies such as **Python, OpenCV, TensorFlow, image processing techniques, and sensor integration**. The expected outcomes include faster inspections, improved defect detection accuracy, optimized repair planning, and reduced maintenance costs.

The system is expected to improve road safety, support infrastructure management, and reduce economic losses caused by poor road conditions.

Keywords: Road Defect Detection, Image Processing, Smart Infrastructure, Road Maintenance, Computer Vision.

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I would also like to thank the **Department of Information Technology of Sri Lanka Institute of Information Technology (SLIIT)** for providing the opportunity and necessary academic resources to conduct this research project.

My appreciation is also extended to my lecturers and research coordinators for their advice and support during the proposal preparation stage. Their feedback and academic guidance helped improve the quality of this research work.

Finally, I would like to express my deepest gratitude to my family and friends for their encouragement, patience, and support during the completion of this proposal.

1. INTRODUCTION

1.1 Background and Context

Road infrastructure is an essential component of modern transportation systems. Well-maintained roads ensure safe travel, efficient transportation of goods, and economic development. However, road surfaces deteriorate over time due to heavy traffic loads, weather conditions, poor construction quality, and lack of regular maintenance. This deterioration results in defects such as potholes, cracks, surface erosion, and uneven road surfaces.

In many countries, including Sri Lanka, road inspection is mainly conducted through manual visual inspections by road authorities and engineers. This process requires significant time and human effort and is highly dependent on human observation and judgment. As a result, many road defects are not detected at an early stage, which leads to further road damage, higher repair costs, and increased risk of road accidents. Early identification of road defects is therefore critical for maintaining road quality and improving transportation safety.

Poor road conditions affect several groups of stakeholders. Drivers and public transport operators face vehicle damage and unsafe driving conditions due to potholes and uneven surfaces. Pedestrians are also exposed to safety risks when roads are damaged. In addition, road maintenance authorities experience difficulties in monitoring large road networks and prioritizing repair activities.

Globally, road damage contributes significantly to traffic accidents and economic losses. According to international road safety reports, poor road conditions increase vehicle maintenance costs and reduce transport efficiency. In developing countries, limited resources and the absence of automated monitoring systems make road maintenance even more challenging.

Several modern approaches have been introduced to improve road inspection. These include camera-based monitoring systems, pavement management systems, and automated road survey vehicles. However, many of these solutions require expensive equipment, specialized vehicles, and advanced sensors, which makes them difficult to implement in developing regions.

To address this challenge, this research proposes a **road defect detection module** that uses image processing techniques to automatically identify road surface defects. The system captures road images using cameras and analyzes them using computer vision algorithms to detect defects such as potholes, cracks, and uneven surfaces. By automating the detection process, the system aims to improve the accuracy and efficiency of road inspections.

This research supports the **United Nations Sustainable Development Goal (SDG) 9 – Industry, Innovation, and Infrastructure**, which focuses on building resilient

infrastructure, and **SDG 11 – Sustainable Cities and Communities**, which promotes safer and more sustainable transportation systems.

The project belongs to the **Technology Integration and Management (TIM)** research cluster because it integrates image processing technologies with infrastructure monitoring and management solutions.

1.2 Literature Review

In recent years, researchers have explored the use of computer vision and machine learning techniques to automatically detect road surface defects. Abbas and Ismael (2021) proposed an automated pavement distress detection system using image processing techniques such as edge detection and image segmentation to identify cracks and potholes. Their study demonstrated that automated image analysis can significantly improve defect detection accuracy compared to traditional manual inspection methods.

Another study by Sivanraj et al. (2025) developed a pothole detection model using the YOLO deep learning algorithm. The model analyzed road images captured by cameras and successfully detected potholes with high accuracy. The research also integrated traffic data to support road maintenance decision-making. However, the system mainly focused on pothole detection and did not address other types of road defects such as cracks or surface erosion.

Several researchers have also applied Convolutional Neural Networks (CNNs) to detect road defects from image datasets. CNN-based models can automatically learn features from road images and classify defects based on their visual patterns. These models have shown promising results in detecting cracks, potholes, and surface irregularities in large image datasets.

In addition to research studies, some commercial road inspection systems have been developed in developed countries. These systems use specialized survey vehicles equipped with high-resolution cameras, LiDAR sensors, and GPS systems to collect detailed road condition data. While these systems provide accurate road quality information, they require expensive hardware and highly specialized equipment.

Despite these advancements, many existing systems still have several limitations. Most research focuses only on detecting a single type of defect, such as potholes or cracks, rather than identifying multiple defect types. In addition, many systems rely on datasets collected from developed countries, which may not represent road conditions in developing regions. Lighting conditions, road materials, and traffic patterns differ significantly between countries, which affects the accuracy of these models when applied locally.

1.3 Research Gap

Although many studies have explored automated road defect detection using computer vision and machine learning techniques, several limitations remain in existing approaches.

First, many existing systems focus only on detecting a single type of defect, such as potholes or cracks. However, real-world road surfaces often contain multiple types of defects simultaneously, including cracks, erosion, and uneven surfaces. A more comprehensive detection approach is required to accurately identify different types of road damage.

Second, several automated road inspection systems require expensive equipment such as LiDAR sensors, specialized vehicles, or advanced scanning technologies. These systems are not practical for developing countries where infrastructure monitoring budgets and technical resources are limited.

Third, many machine learning models are trained using datasets collected in developed countries. Road materials, environmental conditions, and lighting conditions in these datasets may differ from those in Sri Lanka, which reduces the reliability of these models when applied to local road environments.

Therefore, there is a need for a cost-effective and adaptable road defect detection system that can accurately identify multiple types of road surface defects using standard camera images.

This research addresses this gap by developing an **image-processing-based road defect detection module** that can automatically detect potholes, cracks, surface erosion, and uneven road surfaces from captured road images. The system aims to provide a practical and scalable solution for improving road inspection efficiency and supporting better road maintenance planning.

2. OBJECTIVE

2.1 Main Objective

1. To develop an image-processing-based module that can automatically detect and classify road surface defects such as potholes, cracks, surface erosion, and uneven road surfaces from captured road images.

2.2 Specific Objectives

2. To capture high-quality road surface images using camera modules for road defect analysis.
3. To preprocess captured images using image processing techniques such as noise removal, image enhancement, and contrast adjustment to improve detection accuracy.
4. To detect different types of road defects including potholes, cracks, surface erosion, and uneven surfaces using computer vision algorithms.
5. To classify detected defects based on their type and severity using machine learning or image analysis methods.
6. To generate a defect detection output that marks the location and type of road defects for further analysis by the repair estimation modules.

3. METHODOLOGY

Research Approach

This research follows a **Design Science Research approach**, as it focuses on designing and developing a technological artifact that solves a real-world problem. The artifact in this research is an **automated road defect detection module** that uses image processing and computer vision techniques to identify road surface damages.

The approach involves designing the detection system, implementing algorithms for image processing, testing the model using road image datasets, and evaluating its performance based on detection accuracy. Experimental testing will be conducted to analyze the effectiveness of the proposed defect detection method under different road conditions.

Technical Methods and Algorithms

The road defect detection module will use **computer vision and machine learning techniques** to analyze road surface images and identify defects.

First, road images will be captured using camera modules mounted on a vehicle or handheld device. These images will be processed using image preprocessing techniques such as noise reduction, image enhancement, and contrast adjustment to improve image quality.

After preprocessing, **feature extraction techniques** will be applied to identify patterns associated with road damage. Edge detection and texture analysis methods will be used to identify irregularities in the road surface.

To detect and classify defects such as potholes, cracks, erosion, and uneven surfaces, machine learning and deep learning algorithms will be applied. Convolutional Neural Networks (CNN) and object detection models such as **YOLO (You Only Look Once)** will be considered for detecting and classifying defects within the images.

The output of this module will include the **type of defect, its location, and the area affected**, which will be used by the subsequent system modules for repair estimation.

Tools and Platforms with Justification

The system will be developed using **Python**, which is widely used for machine learning and computer vision applications.

Several software libraries and platforms will be used to support the development process. **OpenCV** will be used for image preprocessing and computer vision operations such as edge detection and image segmentation. **TensorFlow or PyTorch** will be used to implement deep learning models for defect detection. These frameworks provide efficient tools for training and testing image classification and object detection models.

The system will be developed and tested using **Jupyter Notebook or Google Colab**, which provide a convenient environment for running machine learning experiments and processing image datasets.

Data Collection and Ethical Considerations

The dataset for this research will be collected using a **hybrid approach**. Primary data will be collected by capturing road images from selected road segments using a camera device. This will ensure that the dataset represents local road conditions, lighting environments, and defect patterns found in Sri Lanka.

In addition, secondary datasets from publicly available road damage image datasets will be used to support model training and improve detection performance.

Since the system captures images in public environments, ethical considerations must be addressed. Any identifiable information such as vehicle license plates or human faces appearing in captured images will be blurred or removed to protect privacy. Necessary permissions will also be obtained if road images are collected from restricted areas.

Validation Strategy and Performance Metrics

The performance of the road defect detection module will be evaluated using experimental testing with labeled road image datasets. The dataset will be divided into **training and testing sets** to train the machine learning model and evaluate its performance.

Several evaluation metrics will be used to measure the effectiveness of the defect detection system. These include **accuracy**, **precision**, **recall**, and **F1-score**, which are commonly used metrics for image classification and object detection tasks.

In addition, detection performance will be evaluated based on the system's ability to correctly identify different types of road defects under various lighting conditions and road surface environments. The results will help determine the reliability and practical usability of the proposed defect detection module.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

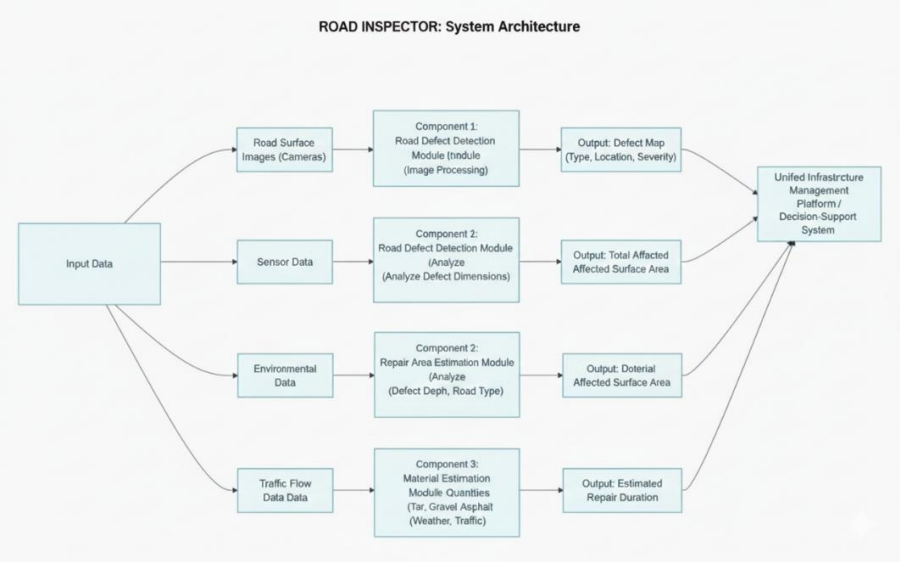
Overview of Proposed Solution and Its Components

The proposed solution for this component is an **image-processing-based road defect detection module** that automatically identifies visible road surface defects from captured images. This module is a core part of the ROAD INSPECTOR system, where road images are collected, processed, and analyzed to detect defects such as potholes, cracks, surface erosion, and uneven surfaces. In the overall project, the system includes multiple modules such as defect detection, repair area estimation, material calculation, and repair time prediction. In

this study, the focus is only on the **Road Defect Detection Module**, which acts as the first stage of the full system workflow. This role is also reflected in the project's system overview and workflow, where images are captured first and then processed to identify defects before later modules use the results.

System Diagram of Components

A simple high-level architecture for this component can be represented as follows:



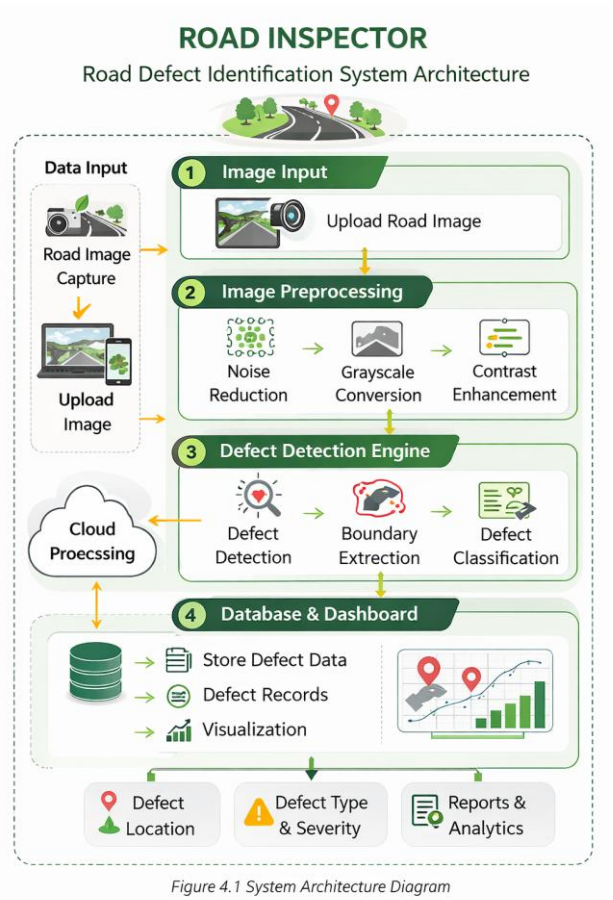


Figure 4.1 System Architecture Diagram

Individual Component and Its Parts

The **Road Defect Detection Module** consists of several subcomponents:

1. Image Capture Subcomponent

This part is responsible for collecting road surface images using a camera module or digital imaging device. The quality of the captured images is important because accurate defect detection depends on clear and detailed road images.

2. Image Preprocessing Subcomponent

The captured images are improved before analysis. This stage includes operations such as noise reduction, brightness adjustment, contrast enhancement, and image resizing. Preprocessing helps reduce unwanted distortions and improves the visibility of road defects.

3. Feature Extraction Subcomponent

At this stage, important visual features are extracted from the image. These features may include edges, texture changes, shape irregularities, and surface discontinuities that are commonly associated with potholes, cracks, and erosion.

4. Defect Detection Subcomponent

This is the main analytical part of the module. Computer vision or deep learning algorithms are applied to detect whether defects are present in the image. This stage identifies possible damaged regions on the road surface.

5. Defect Classification Subcomponent

After detecting damaged regions, the system classifies the defect into categories such as pothole, crack, erosion, or uneven surface.

6. Output Generation Subcomponent

Finally, the module produces an output showing the defect type, location, and marked damaged area. This output can be presented as a defect map or labeled image and passed to other system modules for further processing.

This aligns with the TAF, where defect detection is described as capturing road surface images, preprocessing them, and applying image processing techniques to detect potholes, cracks, uneven surfaces, and tar defects.

Component Interactions

The subcomponents of the road defect detection module interact in a sequential and connected manner.

First, the image capture unit provides the raw road image. This raw image is then passed to the preprocessing unit, which improves the image quality. The processed image is forwarded to the feature extraction stage, where useful visual details are identified. These features are then sent to the defect detection algorithm, which analyzes the image and locates damaged regions. Once a defect is detected, the classification unit determines the defect type. Finally, the results are sent to the output unit, where the detected defects are visualized and stored.

In the broader ROAD INSPECTOR system, the output from this module becomes the input for the repair area estimation module and the other later components. The TAF also describes this relationship by stating that this component produces a defect map showing the type, location, and severity of each detected problem.

Data Flow Explanation

The data flow of this module begins with the collection of road images from real road environments. These images are entered into the system as raw input data.

The raw input images first pass through the preprocessing stage, where the image is cleaned and enhanced. After that, the image features are extracted to identify patterns that may indicate defects. The extracted features are analyzed by the detection algorithm, which identifies damaged road regions. The classification stage then labels each detected region according to the defect type. The final output is a structured result containing information such as defect presence, category, and marked location.

So the data flow can be summarized as:

Raw Road Image → Enhanced Image → Extracted Features → Detected Damage Region → Classified Defect → Final Defect Output

This reflects the workflow described in the uploaded report, where cameras capture road images, image processing algorithms detect defects, and the results are used for later repair estimation.

Scalability and Security Considerations

The proposed architecture is scalable because additional road images can be processed without changing the main structure of the module. The system can also be improved in the future by training the model with a larger dataset, adding more defect categories, or integrating it with mobile devices, drones, or vehicle-mounted cameras. Since the module is designed in separate stages, each subcomponent can be upgraded independently without redesigning the full system.

From a security and privacy perspective, the system should protect any captured public road data. If road images include sensitive information such as vehicle number plates or human faces, those details should be blurred or removed before storing or processing the data.

Access to the stored dataset and output results should be limited to authorized users only. These concerns are also reflected in the TAF, which notes the need for anonymization and permissions during field data collection.

5. USER REQUIREMENTS

Stakeholder Identification

Several stakeholders are involved in the use and benefits of the road defect detection module.

Road maintenance authorities are the primary stakeholders, as they are responsible for monitoring road conditions and planning repair activities. This system will help them detect road defects quickly and accurately.

Engineers and infrastructure inspectors will use the system to analyze road images and identify damaged areas without relying only on manual inspection.

Government transport departments and municipal councils will benefit from improved road monitoring and maintenance planning.

Drivers and road users are indirect stakeholders because accurate detection of road defects helps improve road safety and reduces accidents caused by damaged road surfaces.

Functional Requirements

The road defect detection module should perform the following functions:

1. The system shall capture road surface images using a camera device or imaging system.
2. The system shall preprocess captured images to improve quality by applying techniques such as noise removal, contrast adjustment, and image enhancement.
3. The system shall analyze processed images using computer vision algorithms to detect road defects.
4. The system shall identify different types of road defects such as potholes, cracks, surface erosion, and uneven surfaces.
5. The system shall classify detected defects based on their type and location.
6. The system shall generate an output that marks detected defects on the image for further analysis.
7. The system shall store defect detection results for future inspection and reporting.

Non-Functional Requirements

In addition to functional capabilities, the system must satisfy several non-functional requirements.

Performance:
The system should process road images efficiently and provide defect detection results within a reasonable processing time.

Accuracy:
The detection algorithm should achieve a high level of accuracy in identifying road defects to ensure reliable results.

Usability:
The system interface should be simple and easy to use so that engineers or road inspectors can easily analyze road images.

Scalability:
The system should be capable of handling large numbers of road images as the monitoring area increases.

Security:
Captured road images and system data should be stored securely, and access should be restricted to authorized users.

Reliability:
The system should operate consistently under different road conditions and lighting environments.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the road defect detection system includes **government road authorities, municipal councils, and transportation departments** responsible for road maintenance and infrastructure management. These organizations need efficient tools to monitor road conditions and identify defects early.

Secondary customers include **construction companies, road maintenance contractors, and infrastructure management firms** that are involved in road inspection and repair planning.

The typical customer persona would be a **road maintenance engineer or infrastructure manager** who is responsible for monitoring road conditions, identifying damaged road sections, and planning maintenance activities.

Value Proposition

The proposed road defect detection system offers a **cost-effective and automated solution** for identifying road surface defects using image processing technology. Unlike traditional

manual inspections, the system can detect defects quickly and consistently using captured road images.

The key value provided by the system includes:

- Faster road inspection compared to manual methods
- Improved accuracy in identifying road defects
- Early detection of road damage to reduce repair costs
- Support for data-driven road maintenance decisions

This solution helps road authorities improve road safety while reducing maintenance time and operational costs.

Revenue Model

The system can be commercialized using several revenue models.

One possible approach is a **software licensing model**, where road authorities or infrastructure organizations purchase a license to use the road defect detection software.

Another approach is a **Software-as-a-Service (SaaS) model**, where organizations subscribe to the system and pay a monthly or annual fee for access to the defect detection platform.

The system could also be offered as a **road inspection service**, where a service provider performs road condition analysis for municipalities and infrastructure companies using the developed technology.

Cost Estimation

The development cost of the system mainly includes **software development, dataset collection, model training, and system testing**. Additional costs may include camera devices for image capture and cloud computing resources for processing image data.

Operational costs may include system maintenance, software updates, cloud storage for image datasets, and technical support services.

Pricing Strategy

The pricing strategy should be designed to make the system affordable for infrastructure organizations while maintaining profitability.

Government authorities or municipalities may be offered **institutional pricing or annual licensing plans**. For private companies or infrastructure firms, a **subscription-based pricing model** may be more suitable.

Pricing can also vary depending on the number of roads monitored or the number of images processed by the system.

Competitive Advantage

The proposed system offers several advantages compared to existing road inspection solutions.

First, the system uses **image processing technology**, which allows road defects to be detected automatically without relying solely on manual inspections.

Second, the system is designed to be **cost-effective**, making it suitable for developing countries where expensive road inspection equipment may not be available.

Third, the system can detect **multiple types of road defects** such as potholes, cracks, erosion, and uneven surfaces, improving overall road monitoring capabilities.

Intellectual Property Considerations

The developed road defect detection algorithms and software architecture may be protected through **software copyright or intellectual property rights**. This ensures that the research team retains ownership of the developed technology.

If the system demonstrates strong commercial potential, it may also be possible to apply for **software patents or technology licensing agreements** to protect and commercialize the innovation.

7. BUDGET AND JUSTIFICATION

The following budget estimates the resources required to develop the **road defect detection module** using image processing and machine learning techniques. The module focuses on capturing road images, processing them using computer vision algorithms, and detecting defects such as potholes and cracks.

Item	Estimated Cost (LKR)	Justification
Mobile Camera / Camera Module	12,000	Used to capture road surface images required for defect detection experiments.
External Storage (SSD/HDD)	8,000	Required to store collected road image datasets and trained models.
Cloud GPU Computing	8,000	Used for training deep learning models such as CNN or YOLO for defect detection.
Internet and Dataset Collection	3,000	Used for downloading open-source road datasets and collecting local road images.
Software Tools	0.00	Open-source tools such as Python, OpenCV, TensorFlow, and PyTorch will be used.
Documentation and Printing	2,000	Required for proposal printing, documentation, and research report preparation.
Total Estimated Budget:	**LKR 33,000**	

Justification

The budget focuses only on the resources required to develop the **road defect detection component** of the system. Most software tools used in this research are open-source, which significantly reduces development costs. The main expenses involve image capture hardware, storage for datasets, and cloud computing resources needed for training machine learning models. This ensures that the system can be developed efficiently while maintaining a cost-effective research prototype.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown

The work for the Road Defect Detection component is divided into several stages to ensure systematic development and testing of the module.

Task ID	Task Description
T1	Study existing research on road defect detection and computer vision techniques
T2	Collect road surface image datasets from public sources and local road environments
T3	Design the defect detection module architecture
T4	Perform image preprocessing such as noise removal, contrast enhancement, and image normalization
T5	Develop the defect detection algorithm using image processing and machine learning techniques
T6	Train and test the model using road image datasets
T7	Evaluate detection accuracy using performance metrics
T8	Generate defect detection output and integrate with the system workflow
T9	Documentation and final research report preparation

Timeline Alignment

Phase	Activity
Phase 1	Literature review and dataset collection
Phase 2	System design and architecture planning
Phase 3	Image preprocessing and algorithm development
Phase 4	Model training and defect detection testing
Phase 5	Performance evaluation and system improvements
Phase 6	Documentation and final report preparation

Balanced Workload Distribution

Within the overall research project, each team member is responsible for a different system component. The workload is distributed so that each member focuses on a specific module.

Road Defect Identification using image processing

This division ensures that each component of the system is developed independently while contributing to the overall ROAD INSPECTOR system.

9. GANTT CHART

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Literature Review on Road Defect Detection						
Dataset Collection (Road Images)						
System Architecture Design						
Image Preprocessing Development						
Defect Detection Algorithm Development						
Model Training and Testing						
Performance Evaluation						
Documentation and Report Writing						

Milestones

Milestone	Expected Time
Literature Review Completed	End of Month 2
Dataset Collection Completed	End of Month 4
System Design Completed	End of Month 5
Prototype Defect Detection Model Developed	End of Month 8
Model Training and Optimization Completed	End of Month 10
System Testing and Validation Completed	End of Month 11
Final Research Report and Submission	End of Month 12

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21.APPENDICES

- Survey instruments
- Detailed diagrams
- Preliminary UI mockups
- Risk analysis
- Ethical documents

ROAD INSPECTOR

Automated Road Quality Inspection and Repair Estimation System

R26-IT-023

Project Proposal Report

Wallawita K L T D – IT22252340

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

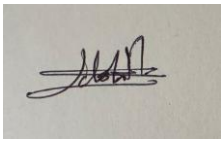
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
WALALLAWITA K L T D	IT22252340	

The above candidate is conducting research for their ı
supervision.

Supervisor Name	Date	Signature
SMB Harshanath		

ABSTRACT

Accurate estimation of road repair areas is essential for effective road maintenance and infrastructure management. In many regions, road repair planning is mainly based on manual measurements and visual inspections carried out by engineers. These traditional methods are time-consuming, prone to human error, and often lead to inaccurate estimation of the damaged road surface. As a result, road authorities may either underestimate or overestimate the repair area, leading to inefficient repair planning, increased maintenance costs, and delays in road restoration.

Although several modern road monitoring systems focus on detecting road defects such as potholes and cracks, many of these systems do not provide accurate calculations of the actual road surface area that requires repair. This creates a gap in road maintenance planning because authorities need precise information about the affected road area in order to estimate resources and repair requirements.

To address this issue, this research proposes a **repair area estimation module** that automatically calculates the damaged road surface area using image analysis techniques. The system receives detected road defects from the road defect detection module and analyzes the boundaries of the damaged regions. Using pixel-based measurement and geometric calculations, the system estimates the real-world surface area that requires repair. The module uses technologies such as **image processing, computer vision algorithms, and geometric calculations** to perform accurate repair area estimation.

The expected outcome of this module is an automated system capable of accurately estimating the total repair area of road defects, which can support better road maintenance planning and resource allocation. By improving the accuracy of repair area estimation, the system can reduce material waste, minimize repair costs, and improve infrastructure maintenance efficiency.

Keywords: *Repair Area Estimation, Road Defect Analysis, Image Processing, Road Maintenance, Infrastructure Monitoring.*

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TABLE OF CONTENTS

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 3
TABLE OF CONTENTS 5
1. INTRODUCTION 7
1.1 Background and Context ..... **Error! Bookmark not defined.**
1.2 Literature Review 7
1.3 Research Gap 8
2. OBJECTIVES 10
2.1 Main Objective 10
2.2 Specific Objectives 10
3. METHODOLOGY 12
3.1 Research Approach 12
3.2 Development Stages 12
3.3 Technologies and Tools 13
3.4 Data Collection Methods 14
4. HIGH-LEVEL SYSTEM ARCHITECTURE 16
4.1 Architecture Overview 16
Data Ingestion Layer 17
Analysis and Prediction Layer 17
Scheduling and Routing Layer 18
Notification and Presentation Layer 18
4.2 System Component Descriptions 18
4.3 Data Flow 19
4.4 Scalability and Security 19
5. USER REQUIREMENTS 20
5.1 Stakeholder Identification 20
5.2 Functional Requirements 20
5.3 Non-Functional Requirements 21
6. COMMERCIALIZATION PLAN 22
6.1 Target Market and Customer Persona 22

6.2 Value Proposition 23
6.3 Revenue Model 23
6.4 Cost Estimation 24
6.5 Pricing Strategy 24
6.6 Competitive Advantage 24
6.7 Intellectual Property Considerations 24
7. BUDGET AND JUSTIFICATION 26
7.1 API Services 26
7.2 Software Licenses 26
7.3 Cloud Services 26
7.4 Human Effort 26
7.5 Overall Budget Summary 26
8. WORK BREAKDOWN STRUCTURE (WBS) 28
9. GANTT CHART 29
10. REFERENCES LIST 30
11. APPENDICES 31
Appendix A – Survey Questionnaire for Road Maintenance Officers 31
Section A: Respondent Background 31
Section B: Current Repair Scheduling Practices 31
Section C: Technology Adoption and Expectations 32
Appendix B – Interview Questions for Road Engineers and RDA Officers 32
Appendix F – Risk Analysis and Mitigation Plan 33
Appendix G – Ethical Consent Form for Data Collection 33

1. INTRODUCTION

1.1 Background and Context

Road infrastructure plays a critical role in ensuring safe transportation, economic development, and efficient mobility of people and goods. In many developing countries, including Sri Lanka, road maintenance activities often face delays due to inefficient inspection processes and poor repair planning. Road surface defects such as potholes, cracks, surface erosion, and uneven pavement surfaces are common problems that significantly reduce road quality and increase the risk of accidents.

Traditionally, road inspection and repair planning are conducted manually by engineers and maintenance officers who visually inspect road conditions and estimate the damaged area. This process is time-consuming, subjective, and often prone to human error. In many cases, inaccurate estimation of the damaged road surface leads to improper planning of repair activities, resulting in material wastage, inefficient resource allocation, and increased maintenance costs.

According to global road safety studies, poor road conditions contribute significantly to traffic accidents, vehicle damage, and transportation inefficiencies. In developing regions, the absence of advanced infrastructure monitoring technologies makes it difficult for road authorities to accurately measure the extent of road damage and prioritize maintenance activities.

Existing road monitoring systems mainly focus on identifying the presence of road defects using cameras, sensors, or mapping technologies. However, many of these systems primarily detect defects without accurately estimating the total repair area required for maintenance operations. As a result, road maintenance teams often rely on manual calculations or rough estimations to determine the size of the affected road surface.

The proposed research introduces an automated approach to **estimate the repair area of detected road defects** using image processing and measurement techniques. The repair area estimation module analyzes processed road images, identifies the boundaries of damaged regions, and calculates the affected surface area using pixel-based and geometric analysis methods.

This research supports the **United Nations Sustainable Development Goals (SDGs)**, particularly **SDG 9 – Industry, Innovation, and Infrastructure**, and **SDG 11 – Sustainable Cities and Communities**, by contributing to the development of safer, more resilient, and efficiently maintained road infrastructure.

Furthermore, this project belongs to the **Technology Integration and Management (TIM)** research cluster because it integrates image processing techniques, measurement algorithms, and data analysis to support intelligent infrastructure management and road maintenance planning.

1.2 Literature Review

Recent research has explored the use of computer vision and machine learning techniques for automated road surface monitoring. Many studies have focused on detecting road defects such as potholes and cracks using image processing algorithms and deep learning models.

Several researchers have applied **Convolutional Neural Networks (CNNs)** and object detection models such as **YOLO** to identify road surface defects with high accuracy. These systems analyze road images captured from cameras and classify different types of defects, enabling faster inspection compared to traditional manual methods.

Other research has explored **image segmentation techniques** to detect the boundaries of road defects. These techniques use edge detection, thresholding, and region segmentation to separate damaged road areas from the surrounding pavement. Identifying the boundaries of defects is an important step that can also support the measurement of damaged areas.

Some studies have attempted to estimate the size of detected defects by analyzing pixel measurements and converting them into real-world dimensions. This approach uses image calibration techniques and geometric calculations to determine the actual surface area affected by road damage.

In addition to academic research, several commercial road inspection systems have been developed using vehicle-mounted platforms equipped with cameras, laser scanners, and GPS technologies. These systems are capable of collecting detailed road condition data, but they often require expensive equipment and specialized vehicles, which makes them less practical for many developing countries.

Although these technologies have improved road monitoring capabilities, many existing systems still focus primarily on detecting defects rather than accurately estimating the repair area required for maintenance operations. This limitation reduces the effectiveness of these systems for infrastructure planning and resource allocation.

1.3 Research Gap

Although numerous studies have proposed automated methods for detecting road defects using image processing and machine learning techniques, most of these approaches focus primarily on defect identification and classification.

Many existing systems lack the capability to accurately estimate the repair area associated with detected road damage. As a result, road authorities often continue to rely on manual measurements or approximate calculations to determine the extent of repair work required.

Furthermore, current road inspection solutions rarely integrate defect detection with repair planning processes such as repair area measurement, material estimation, and maintenance scheduling. The absence of accurate repair area estimation can lead to inefficient repair planning, incorrect resource allocation, and increased maintenance costs.

Another limitation is that many research models are trained using datasets collected in developed countries. Road conditions, materials, and lighting environments in these datasets may differ significantly from those found in developing regions such as Sri Lanka. These differences can reduce the accuracy and reliability of automated road monitoring systems when applied in local environments.

To address these limitations, this research proposes a **repair area estimation module** that calculates the damaged road surface area using pixel-based measurement and geometric analysis techniques applied to processed road images. By accurately measuring the affected surface region, the system can support more efficient road maintenance planning and better allocation of repair resources.

This approach introduces a more practical and integrated solution that complements defect detection by providing accurate measurements of the repair area required for road maintenance activities.

Table 1.2 presents a detailed comparison of existing systems against the proposed module across critical functional dimensions.

Feature	Koch	Maeda	Fan	Arya	Proposed
Road Defect Detection	Yes	Yes	Yes	Yes	**Yes**
Repair Area Estimation	No	Partial	Partial	No	**Yes**
Pixel-Based Area Measurement	No	Yes	Yes	No	**Yes**
Geometric Area Calculation	No	No	Partial	No	**Yes**
Integration with Maintenance Planning	No	No	No	No	**Yes**
Compatibility with Local Road Conditions	No	No	Partial	No	**Yes**

2. OBJECTIVES

2.1 Main Objective

To design and develop a **Repair Area Estimation Module** that, upon receiving detected road defect data, analyzes the boundaries and dimensions of damaged road regions using image processing and geometric measurement techniques to accurately calculate the affected surface area, thereby supporting efficient repair planning and resource allocation within the context of the **ROAD INSPECTOR automated road quality inspection and repair estimation system**.

The objective is:

- **Specific** – automated estimation of damaged road surface area using image-based measurement techniques to support road repair planning in Sri Lankan road maintenance operations
- **Measurable** – accuracy of repair area estimation, pixel-to-real-world measurement conversion accuracy, and system processing time for defect area calculation
- **Achievable** – based on established image processing methods, geometric measurement techniques, and computer vision algorithms for analyzing road surface defects
- **Relevant** – directly supports efficient road maintenance planning, accurate material estimation, and reduction of maintenance cost and resource wastage
- **Time-bound** – within the defined research project duration from **March to November 2026**.

2.2 Specific Objectives

1. To identify the boundaries of detected road defects from processed road images.
2. To measure the size of damaged road regions using pixel-based analysis.
3. To convert pixel measurements into real-world dimensions using scaling or calibration techniques.
4. To calculate the total repair area of the detected road defects using geometric formulas.
5. To provide accurate repair area information to support road maintenance planning and resource allocation.

Table 2.1 summarizes the specific objectives and their expected outcomes.

Specific Objective	Expected Outcome
Collect and preprocess road surface images containing detected defects	Clean and structured image data suitable for repair area estimation
Identify defect boundaries using image segmentation techniques	Accurate detection of damaged road regions
Measure the size of detected defects using pixel-based analysis	Precise pixel-level measurement of damaged areas
Convert pixel measurements into real-world dimensions using calibration methods	Accurate estimation of defect size in square meters
Calculate the total repair area of detected road defects	Reliable data for road repair planning and maintenance
Generate repair area estimation reports for maintenance teams	Improved decision-making and efficient resource allocation

3. METHODOLOGY

3.1 Research Approach

This research follows a **Design Science Research (DSR)** methodology, which focuses on designing, developing, and evaluating innovative technological artifacts to solve real-world problems. In this study, the artifact is the **Repair Area Estimation Module** of the **ROAD INSPECTOR system**.

The purpose of this module is to automatically estimate the damaged road surface area after defects are detected. Accurate repair area estimation supports effective road maintenance planning and resource allocation.

The research process consists of five structured stages: **problem identification, requirements analysis, system design, module development and integration, and performance evaluation**.

3.2 Development Stages

Stage 1 – Requirements Analysis

The system requirements will be identified through literature reviews, discussions with road maintenance engineers, and analysis of current road repair planning practices used by organizations such as the **Sri Lanka Road Development Authority (RDA)** and municipal maintenance departments.

The focus will be on understanding how repair areas are currently estimated, identifying limitations in manual measurement methods, and determining the technical requirements needed to automatically estimate the damaged road surface area.

Stage 2 – System Architecture Design

The architecture of the repair area estimation module will be designed to integrate with the **road defect detection module** of the ROAD INSPECTOR system. The system will receive detected defect locations and boundaries from the defect detection component and perform further analysis to estimate the damaged surface area.

The architecture will include components such as **image preprocessing, defect boundary detection, pixel measurement, and geometric area calculation modules**. Both component-level architecture and data flow diagrams will be developed to illustrate the system structure.

Stage 3 – Data Collection and Preprocessing

Road surface image datasets containing different types of defects such as potholes, cracks, and surface damage will be collected from public datasets and local road environments. These

images will be preprocessed using image enhancement techniques such as noise reduction, grayscale conversion, and edge detection.

The preprocessing stage will prepare the images for accurate boundary detection and measurement of defect regions.

Stage 4 – Area Estimation Algorithm Development

Image processing techniques will be applied to identify the boundaries of detected road defects. The system will measure the size of the damaged region using **pixel-based analysis**.

The measured pixel values will then be converted into real-world measurements using scaling factors or reference calibration methods. **Geometric formulas and pixel-area calculations** will be used to estimate the total repair area required for each defect.

Stage 5 – System Integration and Evaluation

The repair area estimation module will be integrated with the ROAD INSPECTOR system, where it will receive defect detection outputs from **Component 1 (Road Defect Detection Module)**. The module will process these inputs and generate repair area estimations for road maintenance planning.

The performance of the module will be evaluated by comparing the estimated repair areas with manually measured ground truth data. Evaluation metrics such as **measurement accuracy, estimation error rate, and system processing time** will be used to assess the effectiveness of the proposed approach.

3.3 Technologies and Tools

Table 3.1 outlines the selected technologies and tools for module development.

Component	Technology / Tool	Purpose
Road Image Input	Smartphone Camera / Public Road Image Datasets	Capture or obtain road surface images containing defects
Image Preprocessing	Python, OpenCV	Enhance images, remove noise, and prepare images for defect boundary detection
Defect Boundary Detection	OpenCV (Edge Detection, Contour Detection)	Identify and outline the boundaries of detected road defects

Pixel-Based Area Measurement	Python, NumPy	Measure the size of defect regions based on pixel calculations
Area Conversion	Calibration Techniques / Reference Scaling	Convert pixel measurements into real-world measurements (m^2)
Repair Area Calculation	Python Mathematical Functions	Calculate the total damaged road surface area
Backend Processing	Python (Flask / FastAPI)	Store defect information and estimated repair area data
Database	PostgreSQL / MongoDB	Display repair area estimation results for road maintenance planning
Visualization / Dashboard Cloud Hosting	React.js / Chart.js AWS / Google Cloud Platform	Scalable model deployment and data storage Deploy the module and store datasets for scalable processing

3.4 Data Collection Methods

Table 3.2 outlines the data collection methods and sources used in this research.

Data Type	Collection Method	Source
Road surface images	Image capture using mobile cameras and public datasets	Local road environments, public road defect datasets
Detected defect boundaries	Image segmentation results from defect detection module	Component 1 output (Road Defect Detection Module)
Pixel measurement data	Image processing and contour analysis	OpenCV image analysis results
Calibration reference data	Measurement using known road surface dimensions or markers	Field measurements from selected road sections

Ground truth repair measurements	Manual measurement of defect size for validation	Road Development Authority (RDA) / field observations
Historical road damage images	Collection of previously recorded defect images	Public road inspection datasets and maintenance records

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Architecture Overview

The **Repair Area Estimation Module** adopts a layered architecture consisting of four functional layers: the **Data Input Layer, Image Processing Layer, Area Calculation Layer, and Visualization and Integration Layer**. This layered design ensures efficient processing of road images, accurate measurement of damaged regions, and seamless integration with other components of the **ROAD INSPECTOR system**.

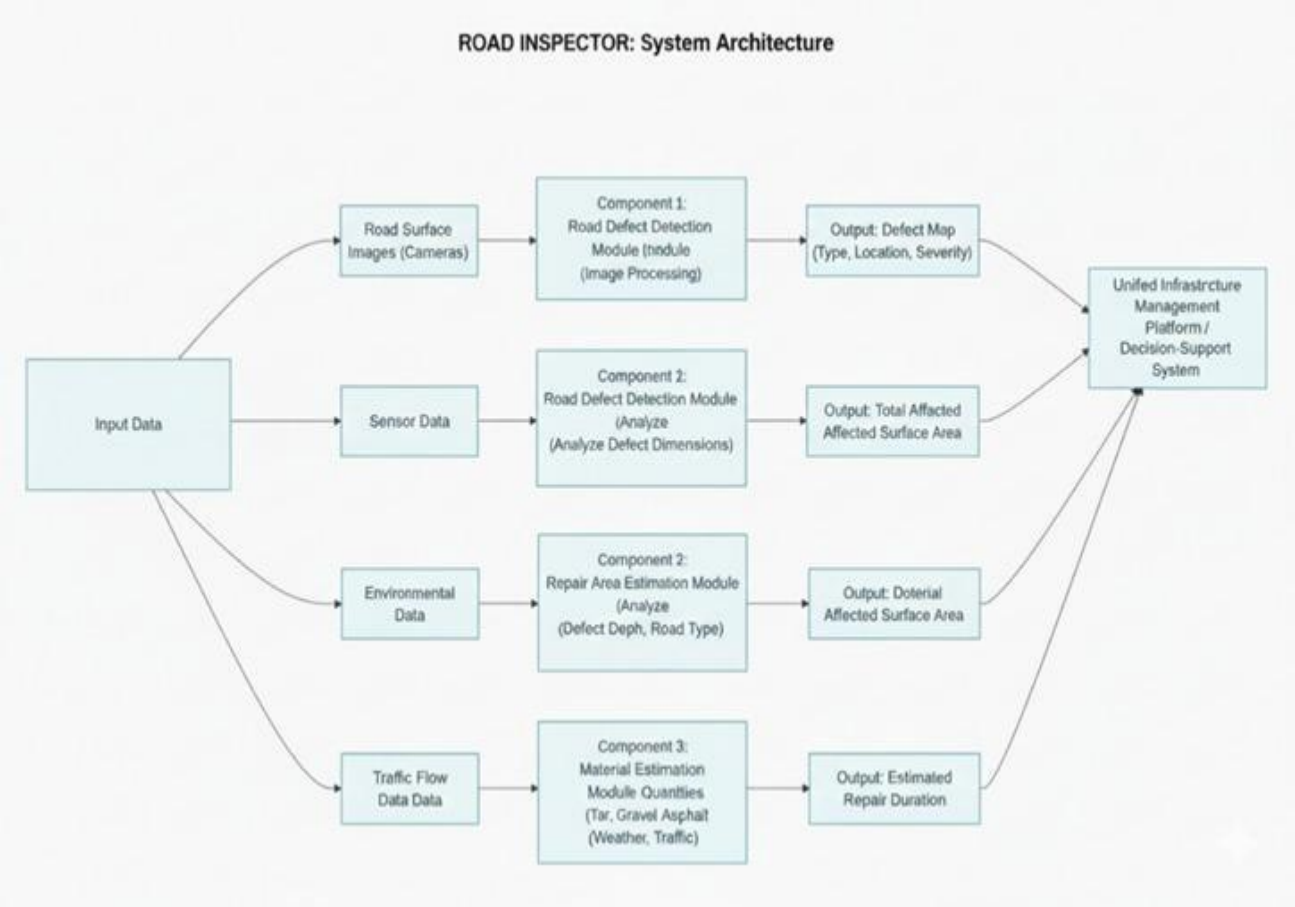




Figure 4.1 System Architecture Diagram

Data Ingestion Layer

- Defect location and boundary data received from Component 1 (Road Defect Detection Module)
- Road surface images captured from cameras or obtained from road defect image datasets
- Pixel measurement data generated through image processing techniques
- Calibration reference data and ground truth measurements collected from road inspection records or field measurements provided by the Sri Lanka Road Development Authority (RDA)

Analysis and Prediction Layer

- Defect boundary detection and contour extraction from processed road imagesTraffic density scoring and peak period identification

- Pixel-based measurement of damaged road regions using image processing techniques
- Surface damage region segmentation using OpenCV image analysis methods
- Conversion of pixel measurements into real-world dimensions using calibration and geometric scaling techniques
- Automated repair area calculation for each detected road defect

Scheduling and Routing Layer

- Repair area calculation engine that computes the damaged road surface area using pixel-based and geometric measurement techniques.
- Conversion of detected defect regions into real-world surface measurements (square meters)
- Generation of repair area data that can be used by the material estimation and repair planning modules

Notification and Presentation Layer

- Display of estimated repair area results on the ROAD INSPECTOR dashboard for maintenance planners
- Visualization of detected defect regions and calculated repair areas on the system interface
- Integration with the ROAD INSPECTOR central operations panel to provide repair area data for material estimation and maintenance planning modules

4.2 System Component Descriptions

Table 4.1 provides detailed descriptions of all key module components.

Component	Description
Image Input Module	Receives road surface images and defect location metadata from the Road Defect Detection Module (Component 1). These images provide the base data required for estimating the damaged road surface area.
Image Preprocessing Engine	Processes the input images using techniques such as noise reduction, grayscale conversion, and contrast enhancement to improve image quality before analysis.
Defect Boundary Detector	Applies image processing algorithms such as edge detection and contour extraction to accurately identify the boundaries of road defects.

Pixel Measurement Processor	Measures the size of detected defect regions in pixel units using image analysis techniques.
Area Conversion Module	Converts pixel-based measurements into real-world dimensions using calibration factors and scaling methods.
Repair Area Calculation Engine	Computes the total damaged road surface area using geometric calculations and processed image data.
Data Storage Module	Stores defect measurements and calculated repair area values in the system database for future analysis and reporting.
Repair Area Visualization Dashboard	Displays estimated repair areas and defect regions through the ROAD INSPECTOR interface to support road maintenance planning and decision-making.

4.3 Data Flow

1. The **Road Defect Detection Module (Component 1)** transmits defect location coordinates, defect boundaries, surface type, and defect metadata to the **Repair Area Estimation Module** via REST API.
2. The **Image Input Module** receives the road surface images and defect boundary information required for further analysis.
3. The **Image Preprocessing Engine** performs image enhancement operations such as noise reduction, grayscale conversion, and contrast adjustment to improve the quality of the input images.
4. The **Defect Boundary Detector** analyzes the processed images and extracts the exact contours of the damaged road regions using image processing algorithms such as edge detection and contour analysis.
5. The **Pixel Measurement Processor** calculates the size of each detected defect region in pixel units based on the extracted boundaries.
6. The **Area Conversion Module** converts pixel-based measurements into real-world dimensions using calibration factors and geometric scaling techniques.
7. The **Repair Area Calculation Engine** computes the total damaged road surface area using geometric formulas and aggregated defect measurements.
8. The **Repair Area Data Output Module** transmits the calculated repair area results to other ROAD INSPECTOR components, including the **Material Requirement Calculation Module and Repair Planning Module**, and displays the results on the system dashboard.

4.4 Scalability and Security

The **Repair Area Estimation Module** is designed for **cloud-based scalability**, enabling the system to process and analyze road surface images from multiple defect locations simultaneously. The module follows a **modular architecture**, where individual components such as image preprocessing, defect boundary detection, and repair area calculation can operate independently. This modular design allows the system to scale efficiently when large volumes of road inspection data are processed across multiple geographic locations.

Cloud deployment ensures that the system can handle increased workloads during large-scale road inspections or infrastructure monitoring campaigns. Horizontal scaling of the backend services enables the platform to maintain consistent performance when multiple repair area estimation requests are processed concurrently.

Security measures are implemented to protect the integrity and confidentiality of system data. All communication between system components and external services is secured through **encrypted API communication using TLS/HTTPS protocols**. Access to the repair area estimation dashboard is controlled through **role-based access control (RBAC)** to ensure that only authorized maintenance personnel and administrators can view or modify system data.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The **Repair Area Estimation Module** serves the following primary stakeholders.

- **Road Maintenance Planners:** Primary users who require accurate estimation of damaged road surface areas to plan repair operations, allocate resources efficiently, and determine the scale of repair work required.
- **Municipal Road Authorities:** Supervisory stakeholders who require visibility into the estimated repair areas of road defects in order to prioritize maintenance activities and manage infrastructure repair budgets effectively.
- **Maintenance Field Teams:** Operational teams who use repair area information to understand the extent of road damage and prepare the necessary materials, equipment, and workforce required for repair tasks.
- **Material Estimation Engineers:** Technical personnel who rely on accurate repair area measurements to calculate the quantity of materials required for road repair operations.

- ROAD INSPECTOR System Administrator: Responsible for managing system configurations, monitoring module performance, maintaining the image processing infrastructure, and ensuring smooth integration between system components.

5.2 Functional Requirements

Table 5.1 details the functional requirements of the proposed module.

ID	Functional Requirement	Stakeholder
FR1	The system shall receive defect location data and defect boundaries from the Road Defect Detection Module.	System Components
FR2	The system shall process road surface images to enhance quality using preprocessing techniques such as noise reduction and contrast adjustment.	System Components
FR3	The system shall detect and extract the boundaries of damaged road regions using image processing algorithms.	Maintenance Teams
FR4	The system shall measure the size of detected defect regions using pixel-based analysis techniques.	Maintenance Engineers
FR5	The system shall convert pixel-based measurements into real-world dimensions using calibration or scaling methods.	Maintenance Engineers
FR6	The system shall calculate the total repair area of detected road defects using geometric calculations.	Maintenance Teams, Planners
FR7	The system shall store calculated repair area data for further analysis and reporting.	Municipal Authorities
FR8	The system shall display estimated repair areas and defect regions on the ROAD INSPECTOR system dashboard.	Road Maintenance Planners

5.3 Non-Functional Requirements

Table 5.2 details the non-functional requirements of the proposed module.

Category	Requirement	Description
Performance	**Image processing time ≤ 5 seconds**	The system must process road images and estimate the repair area within five seconds of receiving defect data.
Accuracy	**Repair area estimation error ≤ 10%**	The estimated repair area must have an error rate of no more than 10% compared to manually measured ground truth values.
Usability	**User-friendly dashboard interface**	The system interface must allow maintenance planners to easily view detected defects and calculated repair areas.
Reliability	**Continuous system availability**	The system must operate reliably and continue processing repair area estimation requests without interruption.
Scalability	**Multi-location processing support**	The architecture must support simultaneous processing of road defect data from multiple locations.
Security	**Encrypted data transmission**	All system communications must use secure protocols such as TLS/HTTPS to protect road inspection data.
Data Integrity	**Secure storage of analysis results**	Repair area estimation results and defect data must be stored securely in the database to prevent data loss or corruption.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the Environmental and Repair Time Analysis Module includes government road maintenance authorities, municipal engineering departments, national highway operators, and private road infrastructure contractors who require data-driven scheduling and disruption management tools.

A typical customer persona would be a road maintenance planning officer in a municipal council or the Sri Lanka Road Development Authority who needs accurate repair duration estimates and

scheduling recommendations to coordinate field teams, manage road closures, and communicate repair timelines to the public.

Table 6.1 summarizes the target market segmentation.

Segment	Target Customer	Use Case
Primary	Municipal road maintenance authorities	Repair scheduling, duration planning, and resource allocation
Secondary	National highway and expressway operators	Large-scale road maintenance coordination and disruption management
Tertiary	Private infrastructure contractors	Project scheduling support and weatherrisk planning
Emerging	Navigation and mapping application providers	Integration of road closure and alternate route data into user applications

6.2 Value Proposition

The proposed module delivers measurable value across multiple dimensions of road maintenance operations.

- Data-driven repair scheduling that replaces subjective manual planning with accurate, ML-powered duration estimates
- Weather-aware scheduling that avoids repair operations during adverse conditions, reducing rework and material wastage
- Traffic-optimized timing that minimizes road closure impact by scheduling repairs during off-peak periods
- Automated alternate route suggestions that reduce road user disruption and improve public communication
- Real-time stakeholder alerts that provide proactive information to drivers, maintenance teams, and municipal authorities

6.3 Revenue Model

The module is offered as part of the broader ROAD INSPECTOR Software-as-a-Service (SaaS) platform. Specific revenue streams include subscription fees for municipal and government road authorities based on the number of active repair sites monitored, integration licensing fees for navigation application providers who incorporate alternate route data, and premium scheduling analytics packages offering historical repair performance reporting and model performance dashboards.

6.4 Cost Estimation

Table 6.2 summarizes the estimated development and operational costs for this module.

Cost Category	Development Cost	Annual Operating Cost
Weather & Traffic API subscriptions	LKR 46,650	LKR 93,300
Software licenses and development tools	LKR 37,320	LKR 24,880
Cloud hosting and ML model deployment	LKR 62,200	LKR 93,300
Human effort (development and engineering)	LKR 310,000	LKR 186,600
Total Estimated Cost	**LKR 456,170**	**LKR 398,080**

6.5 Pricing Strategy

A tiered pricing model is proposed.

- Basic Plan – Small municipal councils with up to ten concurrent repair sites: weather analysis, duration prediction, and notification features
- Standard Plan – Medium authorities with up to thirty sites, including full scheduling engine and alternate route suggestions
- Enterprise Plan – National highway operators and large government departments with unlimited sites, advanced ML reporting, custom alert integrations, and API access for third-party applications

6.6 Competitive Advantage

- First integrated system combining weather analysis, traffic flow, ML duration prediction, and route suggestion within a road maintenance scheduling context
- ML model trained specifically on Sri Lankan road repair data incorporating local surface materials, climate patterns, and traffic behavior
- Proactive scheduling recommendations that prevent repair inefficiency rather than reporting problems after they occur
- Seamless integration with the broader ROAD INSPECTOR platform providing end-toend road inspection and maintenance decision support

6.7 Intellectual Property Considerations

The proposed module may generate intellectual property assets including the trained ML model for repair duration prediction incorporating Sri Lankan road and environmental data, the scheduling optimization algorithm combining weather and traffic signals, and the integrated

alternate route suggestion workflow. The development team may consider software copyright protection, proprietary algorithm documentation, and potential patent registration for the unified environmental-aware scheduling methodology.

7. BUDGET AND JUSTIFICATION

The following budget outlines the estimated costs required to design, develop, and implement the Environmental and Repair Time Analysis Module. The budget covers API subscriptions, software tools, cloud infrastructure, and human effort required for module development and deployment.

7.1 API Services

Weather and traffic API access keys are essential for retrieving real-time and historical environmental and traffic data. OpenWeatherMap provides current conditions, forecasts, and historical weather records. A traffic API such as Google Maps Platform or HERE provides live congestion and speed data. These subscriptions are required both during development for testing and in production for live operation.

7.2 Software Licenses

Development environment licenses include programming IDEs, data analysis tools for model training and validation, machine learning libraries, and UI design software for the scheduling dashboard. These tools are required throughout the full development lifecycle from data preprocessing to deployment.

7.3 Cloud Services

Cloud hosting is required for the backend API, ML inference service, and database storage. A managed cloud platform such as AWS or Google Cloud Platform provides scalable compute resources for real-time prediction and alert dispatch. Additional cloud storage is required for historical repair records, weather and traffic logs, and model artifacts.

7.4 Human Effort

The primary human effort categories include a backend developer for API integration and system implementation, an ML engineer for data preprocessing, model training, and evaluation, a UI/UX designer for the scheduling dashboard, and a QA engineer for end-to-end system testing and performance validation.

7.5 Overall Budget Summary

Table 7.1 presents the detailed proposed budget breakdown.

Category	Description	Estimated Cost
API Services	OpenWeatherMap and traffic API subscription keys for real-time data access	LKR 45,000

ML Libraries & Tools	Machine learning model development, training environment, and evaluation tools	LKR 25,000
Cloud Services	Cloud hosting for backend API, model inference service, and database storage	LKR 65,000
Software Development	IDE licenses, mapping SDK access, and data visualization software	LKR 40,000
Human Effort – Dev	Backend developer for API integration, ML engineer for model development	LKR 220,000
Human Effort – Design/QA	Dashboard UI/UX design, system testing, and quality assurance	LKR 100,000
Total Estimated Budget		**LKR 495,000**

The estimated total budget for developing the Environmental and Repair Time Analysis Module is approximately LKR 495,000. This budget covers essential API subscriptions, software tools, cloud resources, and human expertise required for module implementation. The investment is justified by the system's potential to significantly improve road maintenance scheduling efficiency, reduce repair rework, and minimize disruption to road users across Sri Lanka.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure outlines the complete task decomposition for this research module. Each task corresponds to a deliverable directly supporting the module objectives.

Task	Task Description	Timeline	Deliverable
T1	Conduct literature review on environmental analysis, traffic modeling, and repair scheduling systems	March 2026	Literature review chapter
T2	Collect stakeholder requirements – interviews with RDA officers and road maintenance engineers	March – April 2026	Stakeholder requirements report
T3	Integrate weather data API and implement real-time weather retrieval for defect locations	April 2026	Weather data integration module
T4	Integrate traffic flow API and develop traffic density analysis pipeline	April – May 2026	Traffic analysis module
T5	Collect and preprocess historical repair duration records for ML model training	May 2026	Annotated repair duration dataset
T6	Train and validate ML repair duration prediction model using weather, traffic, and surface features	May – June 2026	Trained prediction model
T7	Evaluate model accuracy and performance using MAPE, RMSE metrics on test dataset	June 2026	Model evaluation report
T8	Develop optimal start time recommendation engine integrating all environmental signals	July 2026	Scheduling engine module
T9	Implement alternate route suggestion module using mapping and directions APIs	July – August 2026	Route suggestion module

T10	Build scheduling dashboard for municipal authorities with maps, timelines, and alerts	August – September 2026	Web dashboard prototype
T11	Integrate module with defect detection and repair estimation components and conduct end-to-end testing	October 2026	Integration test report
T12	Prepare final documentation, evaluation results, and research paper	November 2026	Final research report

9. GANTT CHART

The following Gantt chart illustrates the project milestone timeline for the Environmental and Repair Time Analysis Module, aligned to the 2026 Regular Batch Research Project Plan.

Milestone / Deliverable	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Proposal Submission	●								
Literature Review Complete	●								
Weather & Traffic API Integration		●							
Dataset Collection Complete		●							
Progress Presentation 1			●						
ML Model Training Complete			●						
Model Evaluation Complete				●					
Scheduling Engine Implemented					●				
Route Suggestion Module Ready					●				
Progress Presentation 2						●			
Dashboard Prototype Ready							●		
System Integration Complete								●	

Final Report Submission									●
Final Presentation & Viva									●

Figure 9.1 – Project Gantt Chart Timeline (2026 Regular Batch Research Project Plan)

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11. APPENDICES

Appendix A – Survey Questionnaire for Road Maintenance Officers

Section A: Respondent Background

15. What is your role in road maintenance operations?

- Road Maintenance Engineer
- Planning Officer / Project Manager
- Field Supervisor
- Municipal Council Officer

16. How many years of experience do you have in road maintenance or civil infrastructure?

- Less than 2 years
- 2–5 years
- More than 5 years

Section B: Current Repair Scheduling Practices

17. How is repair scheduling currently determined in your operations?

- Based on field engineer judgment and experience
- Following a fixed maintenance schedule
- Using a digital planning or project management tool
- No formal scheduling process

18. What factors do you currently consider when scheduling a road repair?

- Weather conditions
- Traffic volume and peak hours
- Material and crew availability
- Urgency of defect severity

19. Have delays due to adverse weather or high traffic caused repair rework or extended closures?

- Yes, frequently
- Occasionally

- Rarely
- Never

Section C: Technology Adoption and Expectations

20. Would an AI-powered system that recommends optimal repair timing based on weather and traffic be useful?

- Yes
- No
- Not sure

21. What features would you expect from a smart repair scheduling system? (Select all that apply)

- Real-time weather conditions for repair sites
- Traffic density analysis and peak hour identification
- Automated repair duration estimates
- Optimal start time recommendations
- Alternate route suggestions for road users
- Automated stakeholder notification and alerts

Appendix B – Interview Questions for Road Engineers and RDA Officers

22. How do you currently estimate the time required to complete a road repair of a given size?
23. What environmental or external factors most significantly affect repair efficiency in your experience?
24. Have you ever had to abandon or repeat a repair due to weather or traffic issues? If so, how frequently?
25. How do you currently manage traffic disruption during road repair operations?
26. Are alternate routes provided to road users during road closures, and how are these communicated?
27. Would you be willing to use an automated scheduling tool that recommends start times based on weather and traffic data?
28. What concerns do you have about relying on an AI-based system for repair scheduling recommendations?

Appendix F – Risk Analysis and Mitigation Plan

ID	Risk Description	Impact	Mitigation Strategy
R1	Limited availability of historical road repair duration data from Sri Lankan authorities	**High**	Supplement with expert interviews, structured data collection from RDA, and transfer learning from international datasets
R2	Weather API rate limits or service unavailability during active scheduling sessions	**Medium**	Implement local caching of recent weather data and configure fallback to cached forecast data when live API is unavailable
R3	Traffic data inaccuracy in rural or low-connectivity regions of Sri Lanka	**High**	Combine API data with local sensor feeds and historical traffic patterns; flag lowconfidence estimates to users
R4	ML model underfitting due to high variability in road surface types and defect combinations	**High**	Apply ensemble learning techniques, include surface type as categorical feature, and retrain periodically with new operational data
R5	Road users ignoring alternate route suggestions due to low adoption of the notification system	**Medium**	Partner with navigation app providers (e.g., Waze, Google Maps) for integration; deploy multi-channel alerts (SMS, app, radio)
R6	Privacy concerns related to collection of location and traffic behavioral data	**Medium**	Anonymize all location records, comply with Sri Lankan data protection guidelines, and obtain informed consent from data contributors

Appendix G – Ethical Consent Form for Data Collection

Research Project: ROAD INSPECTOR – Automated Road Quality Inspection and Repair Estimation System

Researcher: IT22252340

Institution: Sri Lanka Institute of Information Technology

Purpose of Research: This study involves the collection of road repair duration records, environmental condition data, and traffic observations from road maintenance operations. The data will be used to train and validate machine learning models for repair duration prediction and scheduling optimization.

Voluntary Participation: Your participation in this study is entirely voluntary. You may withdraw from the study at any time without giving a reason.

Data Usage: All collected data will be anonymized and used strictly for academic research purposes. No personally identifiable information will be shared with third parties. Repair records will be stored securely and used only for model training and validation.

Confidentiality: Your identity and organization details will remain strictly confidential throughout the research process. All data will be stored in encrypted cloud storage accessible only to the research team.

Data Retention: All collected data will be retained for the duration of the research project and disposed of securely upon project completion in compliance with institutional guidelines.

Participant Name	Signature	Date

ROAD INSPECTOR

Automated Road Quality Inspection and Repair Estimation System

R26-IT-023

Project Proposal Report

Bandara T C N – IT22113122

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

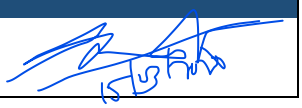
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Bandara T C N	IT22113122	

The above candidate is conducting research for their ı
supervision.

Supervisor Name	Date	Signature
SMB Harshanath		

ABSTRACT

Accurate estimation of road repair requirements is fundamental for resilient infrastructure management. In many regions, including Sri Lanka, road repair planning is predominantly based on manual measurements and subjective visual inspections conducted by field officers. These traditional methods are time-consuming, prone to human error, and frequently lead to inaccurate estimations of the damaged road surface. Consequently, road authorities often either underestimate or overestimate repair needs, resulting in inefficient resource allocation, inflated maintenance costs, and significant delays in road restoration.

While existing modern road monitoring systems have advanced in detecting defects such as potholes and cracks, they often lack the capability to provide accurate calculations of the actual surface area and material volume required for repair. This creates a critical gap in maintenance planning, as authorities require precise, data-driven insights to estimate physical resources and budgets effectively.

To address this issue, this research proposes a **Material Quantity Estimation Module** within the **ROAD INSPECTOR** system. The module automatically calculates the damaged road surface area by analyzing the boundaries of defects identified by the detection component. Utilizing pixel-based measurement and geometric calculations, the system estimates the real-world surface area and integrates sensor-captured depth data to calculate the volume of materials—such as tar, gravel, or asphalt—required for repair. The module leverages technologies including image processing (OpenCV), computer vision algorithms, and sensor fusion to ensure technical accuracy.

The expected outcome is an automated system capable of providing precise repair area and material volume estimations, supporting improved maintenance planning and data-driven resource allocation. By enhancing estimation accuracy, the system aims to minimize material waste, reduce maintenance costs, and significantly improve infrastructure monitoring efficiency.

Keywords: *Repair Area Estimation, Material Quantity Calculation, Road Defect Analysis, Image Processing, Infrastructure Monitoring.*

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Special thanks go to the officers and engineers of the **Sri Lanka Road Development Authority (RDA)** and municipal maintenance departments. Their practical insights helped in understanding the real-world challenges associated with measuring damaged road areas and planning efficient road repair activities. I also acknowledge the professional environment at **Bitumix (Pvt) Ltd**, which has supported my growth within the IT sector and provided context for infrastructure management.

I would also like to acknowledge the collaborative nature of the **ROAD INSPECTOR** project team, whose collective contributions to the overall system architecture helped shape the development of the **Repair Area and Material Estimation** module.

Finally, I express my heartfelt gratitude to my family and friends for their constant encouragement, patience, and moral support throughout this journey. This work would not have been possible without their unwavering motivation.

TABLE OF CONTENTS

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 3
TABLE OF CONTENTS 5
1. INTRODUCTION 7
1.1 Background and Context ..... **Error! Bookmark not defined.**
1.2 Literature Review ..... **Error! Bookmark not defined.**
1.3 Research Gap 9
2. OBJECTIVES 10
2.1 Main Objective 10
2.2 Specific Objectives 10
3. METHODOLOGY 13
3.1 Research Approach 13
3.2 Development Stages 13
3.3 Technologies and Tools 14
3.4 Data Collection Methods 15
4. HIGH-LEVEL SYSTEM ARCHITECTURE 17
4.1 Architecture Overview 17
Data Ingestion Layer ..... **Error! Bookmark not defined.**
Analysis and Prediction Layer ..... **Error! Bookmark not defined.**
Scheduling and Routing Layer ..... **Error! Bookmark not defined.**
Notification and Presentation Layer ..... **Error! Bookmark not defined.**
4.2 System Component Descriptions ..... **Error! Bookmark not defined.**
4.3 Data Flow 21
4.4 Scalability and Security ..... **Error! Bookmark not defined.**
5. USER REQUIREMENTS 21
5.1 Stakeholder Identification 21
5.2 Functional Requirements 22
5.3 Non-Functional Requirements 24

6. COMMERCIALIZATION PLAN 24
6.1 Target Market and Customer Persona 24
6.2 Value Proposition 25
6.3 Revenue Model 25
6.4 Cost Estimation 26
6.5 Pricing Strategy 26
6.6 Competitive Advantage 26
6.7 Intellectual Property Considerations 27
7. BUDGET AND JUSTIFICATION 28
7.1 API Services 28
7.2 Software Licenses 28
7.3 Cloud Services 28
7.4 Human Effort 28
7.5 Overall Budget Summary 29
8. WORK BREAKDOWN STRUCTURE (WBS) 30
9. GANTT CHART 31
10. REFERENCES LIST 32
11. APPENDICES 33
Appendix A – Survey Questionnaire for Road Maintenance Officers 33
Section A: Respondent Background 33
Section B: Current Repair Scheduling Practices 33
Section C: Technology Adoption and Expectations 34
Appendix B – Interview Questions for Road Engineers and RDA Officers 34
Appendix F – Risk Analysis and Mitigation Plan 35
Appendix G – Ethical Consent Form for Data Collection 35

1. INTRODUCTION

1.1 Background and Context

Road infrastructure plays a critical role in ensuring safe transportation, economic development, and efficient mobility of people and goods. In many developing countries, including Sri Lanka, road maintenance activities often face delays due to inefficient inspection processes and poor repair planning. Road surface defects such as potholes, cracks, surface erosion, and uneven pavement surfaces are common problems that significantly reduce road quality and increase the risk of accidents.

A major challenge in road maintenance is the **accurate estimation of materials required for repair work**. Road repair operations typically involve materials such as asphalt, tar mixtures, gravel, aggregates, and concrete depending on the type of road surface. Estimating the correct quantity of these materials is essential for efficient repair planning, cost management, and minimizing material wastage.

Traditionally, the calculation of road repair materials is performed manually by engineers and maintenance officers after visually inspecting the damaged road areas. These professionals estimate the repair area and depth of the defect, and then calculate the required material quantities using manual formulas or rough approximations. This process is time-consuming, highly dependent on human judgment, and often leads to inaccurate results.

In many cases, incorrect material estimation results in either insufficient materials or excessive resource allocation. Insufficient materials can delay repair work and increase labor costs, while excessive material allocation leads to wastage and unnecessary financial expenditure. Therefore, accurate and automated material estimation is essential for improving road maintenance efficiency and reducing operational costs.

Existing road monitoring systems mainly focus on detecting defects such as potholes or cracks using cameras, sensors, or computer vision techniques. However, most of these systems do not extend their functionality to **automatically calculate the quantity of repair materials required** based on the detected damage. As a result, maintenance teams still rely on manual calculations after defect detection, which reduces the overall efficiency of automated inspection systems.

The proposed research introduces an automated **Material Requirement Estimation Module** that calculates the quantity of materials required for road repairs using data obtained from the defect detection and repair area estimation modules. By analyzing the size, depth, and type of road damage, the system will estimate the volume of repair materials required and convert it into practical material quantities based on standard engineering density values.

This automated approach will help road authorities and maintenance teams prepare accurate repair plans, optimize resource utilization, and minimize material wastage during road maintenance operations.

This research also supports the **United Nations Sustainable Development Goals (SDGs)**, particularly **SDG 9 – Industry, Innovation, and Infrastructure**, and **SDG 11 – Sustainable Cities and Communities**, by promoting efficient infrastructure maintenance and sustainable resource management.

Furthermore, this project belongs to the **Technology Integration and Management (TIM)** research cluster because it integrates image processing, measurement algorithms, and engineering material estimation techniques to support intelligent road maintenance planning and infrastructure management.

1.2 Literature Review

Road maintenance and repair planning have been widely studied in the fields of civil engineering, transportation engineering, and computer vision. Traditional road maintenance systems primarily rely on manual inspection methods, where engineers visually examine road surfaces and estimate the level of damage and required repair resources. Although these methods have been used for many years, they often lead to inaccurate assessments due to human error and subjective judgment. As a result, researchers have increasingly focused on developing automated systems for road defect detection and maintenance planning.

Recent studies have applied image processing and machine learning techniques to detect road surface defects such as potholes, cracks, and surface deterioration. For example, Abbas and Ismael (2021) proposed an automated pavement distress detection system using image processing techniques. Their research demonstrated that computer vision algorithms can effectively identify road surface damages by analyzing image features such as edges, textures, and irregular patterns. While this approach improves defect detection accuracy, the study mainly focuses on identifying defects rather than estimating the required repair materials.

Another research study by Sivanraj et al. (2025) introduced a deep learning model based on YOLO for pothole detection and traffic-aware maintenance planning. The system integrates traffic data to prioritize road repairs and improve maintenance scheduling. Although the system provides valuable insights for road maintenance decision-making, it does not include a detailed mechanism for calculating the quantity of repair materials needed for fixing the detected defects.

Several studies have also explored the use of image segmentation and geometric analysis for measuring the dimensions of road damage. Image segmentation techniques can identify the boundaries of potholes and cracks, allowing researchers to calculate the affected surface area using pixel-based measurements. However, most existing systems stop at calculating the defect area and do not extend the analysis to estimate the material volume required for road repair operations.

In the field of civil engineering, the calculation of road repair materials typically involves determining the repair area, defect depth, and material density. Engineers use these parameters to calculate the required volume and weight of materials such as asphalt, gravel, and concrete. However, these calculations are usually performed manually after field inspection, which makes the process slow and prone to estimation errors.

Recent research trends highlight the need for integrated road inspection systems that combine defect detection, area measurement, and material estimation within a single platform. Such systems can significantly improve the efficiency of road maintenance planning by providing accurate, data-driven estimates of repair materials and costs.

Therefore, the proposed research aims to address this limitation by developing an automated **Material Requirement Estimation Module** that calculates the quantity of repair materials based on defect size, depth, and road type. By integrating computer vision outputs with engineering calculation models, the system can provide more accurate and reliable material estimates for road maintenance operations.

1.3 Research Gap

Although numerous studies have proposed automated methods for detecting road defects using image processing and machine learning techniques, most of these approaches primarily focus on defect identification and classification. While these systems are capable of identifying potholes, cracks, and other road surface damages, they rarely extend their functionality to support detailed repair planning processes.

One of the key limitations of existing systems is the lack of automated mechanisms for **estimating the quantity of materials required for road repair**. In many road maintenance operations, engineers must manually calculate the materials needed for repair work after identifying the damaged areas. These calculations are usually based on the estimated repair area, defect depth, and the type of road surface. Because this process is performed manually, it often leads to inaccurate estimations, delays in maintenance planning, and inefficient resource allocation.

Furthermore, most current road inspection solutions do not integrate defect detection outputs with engineering-based material estimation models. Even when automated systems successfully detect road damage, maintenance teams must still rely on manual calculations to determine the required quantities of asphalt, tar mixtures, gravel, or concrete. This separation between defect detection and material estimation reduces the overall efficiency of automated road inspection systems.

Another limitation is that many existing research models are developed using datasets and road conditions from developed countries. The road construction materials, environmental conditions, and damage patterns in these datasets may differ significantly from those found in developing regions such as Sri Lanka. As a result, existing systems may not accurately support material estimation for local road maintenance operations.

To address these limitations, this research proposes an automated **Material Requirement Estimation Module** that calculates the quantity of materials needed for road repair based on the

detected defect size, repair area, and defect depth. The system will use measurement data obtained from processed road images and apply engineering formulas to estimate the volume and weight of repair materials required.

By integrating defect detection outputs with automated material calculation techniques, the proposed system can support more accurate repair planning, reduce material wastage, and improve the efficiency of road maintenance operations.

Table 1.2 presents a comparison of existing systems and the proposed module across key functional capabilities related to material estimation and maintenance planning.

Feature	Koch	Maeda	Fan	Arya	Proposed
Road Defect Detection	Yes	Yes	Yes	Yes	**Yes**
Repair Area Estimation	No	Partial	Partial	No	**Yes**
Pixel-Based Area Measurement	No	Yes	Yes	No	**Yes**
Geometric Area Calculation	No	No	Partial	No	**Yes**
Integration with Maintenance Planning	No	No	No	No	**Yes**
Compatibility with Local Road Conditions	No	No	Partial	No	**Yes**
Material Requirement Estimation	No	No	No	No	**Yes**

2. OBJECTIVES

2.1 Main Objective

To design and develop a **Material Requirement Estimation Module** that, upon receiving detected road defect data, calculates the volume and quantity of repair materials such as asphalt, tar mixtures, gravel, or concrete using defect dimensions, estimated repair depth, and road type,

thereby supporting efficient resource allocation, cost-effective planning, and timely execution of road maintenance within the context of the **ROAD INSPECTOR** automated road quality inspection and repair estimation system.

The objective is:

- **Specific** – automated calculation of repair material quantities based on defect area, depth, and road surface type to support accurate road maintenance planning in Sri Lanka.
- **Measurable** – precision of material volume and weight estimation, accuracy of defect depth measurement, and processing time for material calculation.
- **Achievable** – based on established geometric formulas, civil engineering material standards, and outputs from the defect detection and repair area estimation modules.
- **Relevant** – directly supports efficient road maintenance, minimizes material wastage, and reduces operational costs.
- **Time-bound** – achievable within the defined research project duration from March to November 2026.

2.2 Specific Objectives

1. To receive defect dimensions and estimated repair areas from the road defect detection and repair area estimation modules.
2. To determine the depth of each detected road defect using sensor data or image-based estimation.
3. To calculate the required volume of repair materials based on defect area and depth.
4. To convert material volume into weight using standard density values for different road repair materials.
5. To generate accurate material requirement reports to support road maintenance planning and resource allocation.
6. To integrate the material estimation results into the ROAD INSPECTOR system for automated maintenance decision-making.

Table 2.1 summarizes the specific objectives and their expected outcomes.

Specific Objective	Expected Outcome
Receive defect dimensions and repair area data	Accurate input data from previous modules for material calculation
Determine defect depth using sensor/image data	Precise estimation of repair depth
Calculate volume of repair materials based on defect size	Reliable material quantity in cubic meters

Generate material requirement reports	Improved maintenance planning and resource allocation
Integrate results into ROAD INSPECTOR system	Automated material estimation supporting decision-making
Generate repair area estimation reports for maintenance teams	Improved decision-making and efficient resource allocation

3. METHODOLOGY

3.1 Research Approach

This research follows a **Design Science Research (DSR)** methodology, which focuses on designing, developing, and evaluating innovative technological artifacts to solve real-world problems. In this study, the artifact is the **Repair Area Estimation Module** of the **ROAD INSPECTOR system**.

The purpose of this module is to automatically estimate the damaged road surface area after defects are detected. Accurate repair area estimation supports effective road maintenance planning and resource allocation.

The research process consists of five structured stages: **problem identification, requirements analysis, system design, module development and integration, and performance evaluation**.

3.2 Development Stages

Stage 1 – Requirements Analysis

The system requirements will be identified through literature reviews, discussions with road maintenance engineers, and analysis of current road repair planning practices used by organizations such as the **Sri Lanka Road Development Authority (RDA)** and municipal maintenance departments.

The focus will be on understanding how repair areas are currently estimated, identifying limitations in manual measurement methods, and determining the technical requirements needed to automatically estimate the damaged road surface area.

Stage 2 – System Architecture Design

The architecture of the repair area estimation module will be designed to integrate with the **road defect detection module** of the ROAD INSPECTOR system. The system will receive detected defect locations and boundaries from the defect detection component and perform further analysis to estimate the damaged surface area.

The architecture will include components such as **image preprocessing, defect boundary detection, pixel measurement, and geometric area calculation modules**. Both component-level architecture and data flow diagrams will be developed to illustrate the system structure.

Stage 3 – Data Collection and Preprocessing

Road surface image datasets containing different types of defects such as potholes, cracks, and surface damage will be collected from public datasets and local road environments. These

images will be preprocessed using image enhancement techniques such as noise reduction, grayscale conversion, and edge detection.

The preprocessing stage will prepare the images for accurate boundary detection and measurement of defect regions.

Stage 4 – Area Estimation Algorithm Development

Image processing techniques will be applied to identify the boundaries of detected road defects. The system will measure the size of the damaged region using **pixel-based analysis**.

The measured pixel values will then be converted into real-world measurements using scaling factors or reference calibration methods. **Geometric formulas and pixel-area calculations** will be used to estimate the total repair area required for each defect.

Stage 5 – System Integration and Evaluation

The repair area estimation module will be integrated with the ROAD INSPECTOR system, where it will receive defect detection outputs from **Component 1 (Road Defect Detection Module)**. The module will process these inputs and generate repair area estimations for road maintenance planning.

The performance of the module will be evaluated by comparing the estimated repair areas with manually measured ground truth data. Evaluation metrics such as **measurement accuracy, estimation error rate, and system processing time** will be used to assess the effectiveness of the proposed approach.

3.3 Technologies and Tools

Table 3.1 outlines the selected technologies and tools for module development.

Component	Technology / Tool	Purpose
Road Image Input	Smartphone Camera / Public Road Image Datasets	Capture or obtain road surface images containing defects
Image Preprocessing	Python, OpenCV	Enhance images, remove noise, and prepare images for defect boundary detection
Defect Boundary Detection	OpenCV (Edge Detection, Contour Detection)	Identify and outline the boundaries of detected road defects

Pixel-Based Area Measurement	Python, NumPy	Measure the size of defect regions based on pixel calculations
Area Conversion	Calibration Techniques / Reference Scaling	Convert pixel measurements into real-world measurements (m^2)
Repair Area Calculation	Python Mathematical Functions	Calculate the total damaged road surface area
Backend Processing	Python (Flask / FastAPI)	Store defect information and estimated repair area data
Database	PostgreSQL / MongoDB	Display repair area estimation results for road maintenance planning
Visualization / Dashboard Cloud Hosting	React.js / Chart.js AWS / Google Cloud Platform	Scalable model deployment and data storage Deploy the module and store datasets for scalable processing

3.4 Data Collection Methods

Table 3.2 outlines the data collection methods and sources used in this research.

Data Type	Collection Method	Source
Road surface images	Image capture using mobile cameras and public datasets	Local road environments, public road defect datasets
Detected defect boundaries	Image segmentation results from defect detection module	Component 1 output (Road Defect Detection Module)
Pixel measurement data	Image processing and contour analysis	OpenCV image analysis results
Calibration reference data	Measurement using known road surface dimensions or markers	Field measurements from selected road sections

Ground truth repair measurements	Manual measurement of defect size for validation	Road Development Authority (RDA) / field observations
Historical road damage images	Collection of previously recorded defect images	Public road inspection datasets and maintenance records

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Architecture Overview

The **Material Requirement Calculation Module** adopts a layered architecture consisting of four functional layers: the **Data Input Layer, Material Analysis Layer, Quantity Calculation Layer, and Visualization and Integration Layer**. This layered design ensures efficient processing of repair area and defect data, accurate estimation of required road repair materials, and seamless integration with other components of the **ROAD INSPECTOR system**.

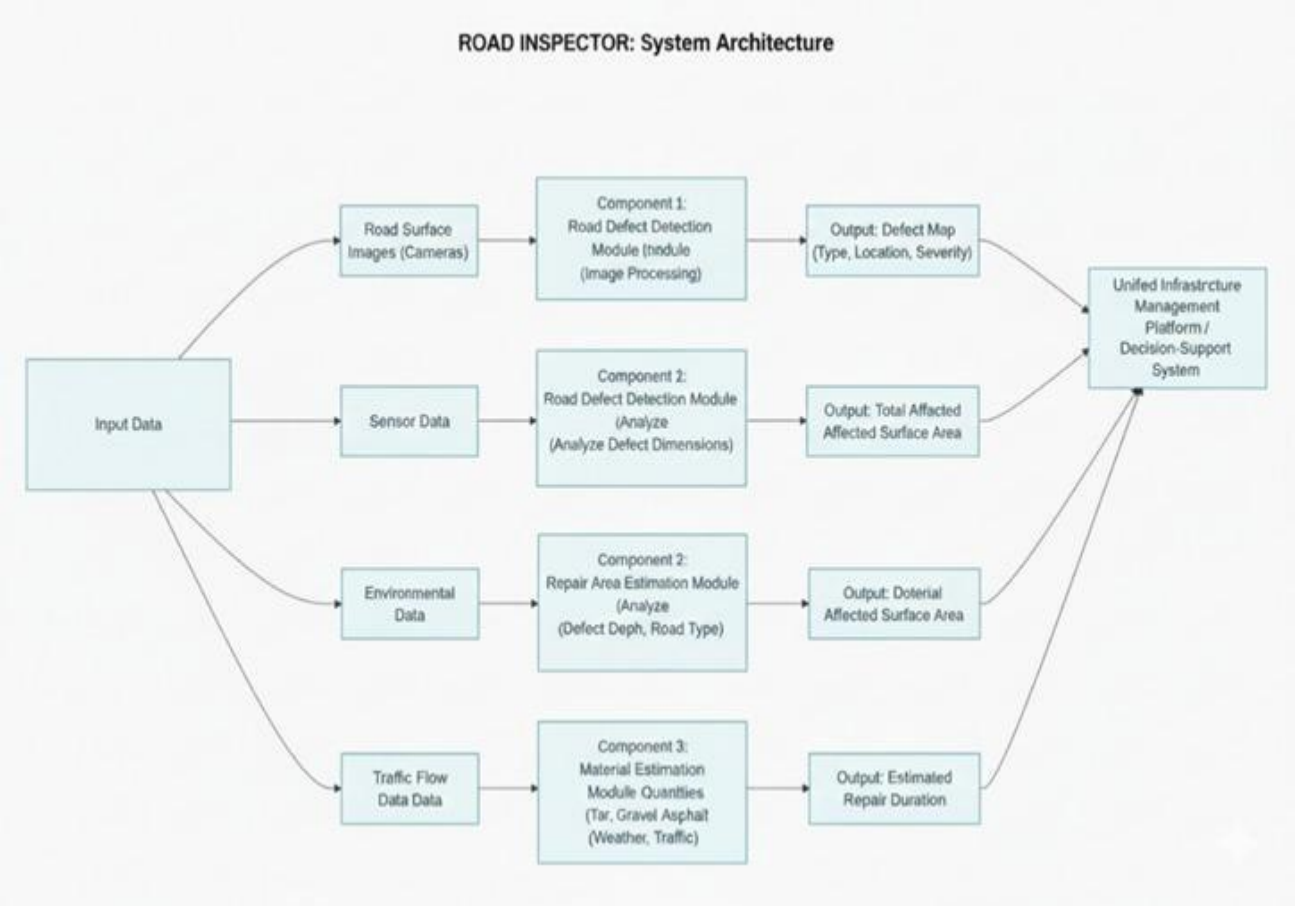




Figure 4.1 System Architecture Diagram

Data Ingestion Layer

• Repair area data and defect measurements received from the Repair Area Estimation Module

• Road defect classification and defect type information obtained from the Road Defect Detection Module

• Road surface condition data and repair specifications obtained from road maintenance datasets or inspection records

•Material specification data and standard repair material ratios collected from **road maintenance guidelines and Road Development Authority (RDA) repair standards**

Layer

- Identification of road defect types and repair categories based on detected road damage data
- Analysis of estimated repair area and severity levels received from the Repair Area Estimation Module
- Classification of suitable repair methods (patching, resurfacing, crack sealing, or reconstruction) based on defect characteristics
- Selection of appropriate repair materials using predefined road maintenance standards and material specification datasets
- Preparation of structured repair parameters required for accurate material quantity calculationScheduling and Routing Layer

Notification and Presentation Layer

- **Display of estimated material requirement results on the ROAD INSPECTOR dashboard for maintenance planners and engineers**
- Visualization of required material quantities, repair material types, and allocation details on the system interface
- Integration with the ROAD INSPECTOR central operations panel to provide material requirement data for repair planning, budgeting, and maintenance scheduling modules

4.2 System Component Descriptions

Table 4.1 provides detailed descriptions of all key module components.

Component	Description
Material Data Input Module	Receives repair area measurements and defect information from the Repair Area Estimation Module. These inputs provide the base data required to estimate the quantity of materials needed for road repair activities.
Image Repair Type Identification Engine	Analyzes defect type, severity, and repair area information to determine the appropriate repair method such as patching, resurfacing, or crack sealing.
Material Specification Processor	Retrieves standard repair material specifications and ratios from road maintenance guidelines and material databases used for road construction and repair.
Material Conversion Module	Converts calculated material quantities into standardized measurement units such as kilograms, tons, or cubic meters for practical usage in road maintenance operations.
Material Requirement Database Module	Stores calculated material requirements and repair material records in the system database for future reference, reporting, and maintenance planning.
Material Requirement Visualization Dashboard	Displays estimated material requirements and repair planning information through the ROAD INSPECTOR dashboard to support decision-making by maintenance planners and engineers.

4.3 Data Flow

1. The **Repair Area Estimation Module** transmits calculated repair area data, defect type, surface type, and defect metadata to the **Material Requirement Calculation Module** via REST API.
2. The **Material Data Input Module** receives repair area measurements and defect information required for estimating the quantity of materials needed for road repair.
3. The **Repair Type Identification Engine** analyzes the defect type, severity level, and surface characteristics to determine the appropriate repair method such as patching, resurfacing, or crack sealing.
4. The **Material Specification Processor** retrieves standard repair material specifications and mix ratios from the road maintenance guidelines and material specification database.

5. The **Material Quantity Calculation Engine** calculates the quantity of required materials such as asphalt mix, cement, aggregates, and fillers based on the estimated repair area and selected repair method.
6. The **Area Conversion Module** converts pixel-based measurements into real-world dimensions using calibration factors and geometric scaling techniques.
7. The **Repair Area Calculation Engine** computes the total damaged road surface area using geometric formulas and aggregated defect measurements.
8. The **Repair Area Data Output Module** transmits the calculated repair area results to other ROAD INSPECTOR components, including the **Material Requirement Calculation Module and Repair Planning Module**, and displays the results on the system dashboard.

4.4 Scalability and Security

The **Material Requirement Calculation Module** is designed for cloud-based scalability, enabling the system to process and analyze repair area data and defect information from multiple road locations simultaneously. The module follows a modular architecture, where individual components such as repair type identification, material specification processing, and material quantity calculation can operate independently. This modular design allows the system to scale efficiently when large volumes of road inspection and repair data are processed across multiple geographic locations.

Cloud deployment ensures that the system can handle increased workloads during large-scale road maintenance operations or infrastructure monitoring activities. Horizontal scaling of the backend services enables the platform to maintain consistent performance when multiple material requirement estimation requests are processed concurrently.

Security measures are implemented to protect the integrity and confidentiality of system data. All communication between system components and external services is secured through encrypted API communication using TLS/HTTPS protocols. Access to the material requirement calculation dashboard is controlled through role-based access control (RBAC) to ensure that only authorized maintenance personnel, engineers, and administrators can view or modify system data.

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5. USER REQUIREMENTS

5.1 Stakeholder Identification

The **Material Requirement Calculation Module serves the following primary stakeholders.**

- Road Maintenance Planners: Primary users who require accurate estimation of repair materials to plan road maintenance operations, allocate resources efficiently, and schedule repair activities effectively.

    - Municipal Road Authorities: Supervisory stakeholders who require visibility into estimated material requirements for road repairs in order to manage infrastructure maintenance budgets and prioritize repair projects.
    - Maintenance Field Teams: Operational teams who use calculated material requirement information to prepare the necessary materials, tools, and equipment needed for road repair operations.

- Material Estimation Engineers: Technical personnel who rely on accurate material requirement calculations to determine the quantity of asphalt, aggregates, cement, and other materials required for road maintenance and rehabilitation.

- ROAD INSPECTOR System Administrator: Responsible for managing system configurations, monitoring module performance, maintaining the image processing infrastructure, and ensuring smooth integration between system components.

5.2 Functional Requirements

Table 5.1 details the functional requirements of the proposed module.

ID	Functional Requirement	Stakeholder
FR1	The system shall receive repair area data and defect information from the Repair Area Estimation Module.	System Components
FR2	The system shall identify the type of road defect and determine the appropriate repair method based on defect characteristics and road surface conditions.	System Components
FR3	The system shall retrieve standard repair material specifications and mix ratios from the material specification database.	Maintenance Teams
FR4	The system shall calculate the quantity of materials required for road repair operations based on the estimated repair area.	Maintenance Engineers
FR5	The system shall convert calculated material quantities into standardized units such as kilograms, tons, or cubic meters.	Maintenance Engineers
FR6	The system shall generate total material requirement estimates for each detected road defect and repair task.	Maintenance Teams, Planners
FR7	The system shall store calculated material requirement data in the system database for future analysis and reporting.	Municipal Authorities
FR8	The system shall display estimated material requirements and repair material details on the ROAD INSPECTOR system dashboard.	Road Maintenance Planners

5.3 Non-Functional Requirements

Table 5.2 details the non-functional requirements of the proposed module.

Category	Requirement	Description
Performance	**Material calculation time ≤ 5 seconds**	The system must calculate required repair materials within five seconds after receiving repair area and defect data.
Accuracy	**Material estimation error ≤ 10%**	The estimated material quantities must have an error rate of no more than 10% compared to standard manual material estimation methods.
Usability	**User-friendly dashboard interface**	The system interface must allow maintenance planners and engineers to easily view calculated material quantities and repair planning information.
Reliability	**Continuous system availability**	The system must operate reliably and continue processing material requirement estimation requests without interruption.
Scalability	**Multi-location processing support**	The architecture must support simultaneous material requirement calculations for road repair tasks across multiple locations.
Security	**Encrypted data transmission**	All system communications must use secure protocols such as TLS/HTTPS to protect road maintenance and repair planning data.
Data Integrity	**Secure storage of material estimation results**	Calculated material requirement data must be stored securely in the database to prevent data loss or corruption.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the Environmental and Repair Time Analysis Module includes government road maintenance authorities, municipal engineering departments, national highway operators, and private road infrastructure contractors who require data-driven scheduling and disruption management tools.

A typical customer persona would be a road maintenance planning officer in a municipal council or the Sri Lanka Road Development Authority who needs accurate repair duration estimates and scheduling recommendations to coordinate field teams, manage road closures, and communicate repair timelines to the public.

Table 6.1 summarizes the target market segmentation.

Segment	Target Customer	Use Case
Primary	Municipal road maintenance authorities	Repair scheduling, duration planning, and resource allocation
Secondary	National highway and expressway operators	Large-scale road maintenance coordination and disruption management
Tertiary	Private infrastructure contractors	Project scheduling support and weatherrisk planning
Emerging	Navigation and mapping application providers	Integration of road closure and alternate route data into user applications

6.2 Value Proposition

The proposed module delivers measurable value across multiple dimensions of road maintenance operations.

- Data-driven repair scheduling that replaces subjective manual planning with accurate, ML-powered duration estimates
- Weather-aware scheduling that avoids repair operations during adverse conditions, reducing rework and material wastage
- Traffic-optimized timing that minimizes road closure impact by scheduling repairs during off-peak periods
- Automated alternate route suggestions that reduce road user disruption and improve public communication
- Real-time stakeholder alerts that provide proactive information to drivers, maintenance teams, and municipal authorities

6.3 Revenue Model

The module is offered as part of the broader ROAD INSPECTOR Software-as-a-Service (SaaS) platform. Specific revenue streams include subscription fees for municipal and government road authorities based on the number of active repair sites monitored, integration licensing fees for navigation application providers who incorporate alternate route data, and premium scheduling

analytics packages offering historical repair performance reporting and model performance dashboards.

6.4 Cost Estimation

Table 6.2 summarizes the estimated development and operational costs for this module.

Cost Category	Development Cost	Annual Operating Cost
Weather & Traffic API subscriptions	LKR 46,650	LKR 93,300
Software licenses and development tools	LKR 37,320	LKR 24,880
Cloud hosting and ML model deployment	LKR 62,200	LKR 93,300
Human effort (development and engineering)	LKR 310,000	LKR 186,600
Total Estimated Cost	**LKR 456,170**	**LKR 398,080**

6.5 Pricing Strategy

A tiered pricing model is proposed.

- Basic Plan – Small municipal councils with up to ten concurrent repair sites: weather analysis, duration prediction, and notification features
- Standard Plan – Medium authorities with up to thirty sites, including full scheduling engine and alternate route suggestions
- Enterprise Plan – National highway operators and large government departments with unlimited sites, advanced ML reporting, custom alert integrations, and API access for third-party applications

6.6 Competitive Advantage

- First integrated system combining weather analysis, traffic flow, ML duration prediction, and route suggestion within a road maintenance scheduling context
- ML model trained specifically on Sri Lankan road repair data incorporating local surface materials, climate patterns, and traffic behavior
- Proactive scheduling recommendations that prevent repair inefficiency rather than reporting problems after they occur
- Seamless integration with the broader ROAD INSPECTOR platform providing end-toend road inspection and maintenance decision support

6.7 Intellectual Property Considerations

The proposed module may generate intellectual property assets including the trained ML model for repair duration prediction incorporating Sri Lankan road and environmental data, the scheduling optimization algorithm combining weather and traffic signals, and the integrated alternate route suggestion workflow. The development team may consider software copyright protection, proprietary algorithm documentation, and potential patent registration for the unified environmental-aware scheduling methodology.

7. BUDGET AND JUSTIFICATION

The following budget outlines the estimated costs required to design, develop, and implement the Environmental and Repair Time Analysis Module. The budget covers API subscriptions, software tools, cloud infrastructure, and human effort required for module development and deployment.

7.1 API Services

Weather and traffic API access keys are essential for retrieving real-time and historical environmental and traffic data. OpenWeatherMap provides current conditions, forecasts, and historical weather records. A traffic API such as Google Maps Platform or HERE provides live congestion and speed data. These subscriptions are required both during development for testing and in production for live operation.

7.2 Software Licenses

Development environment licenses include programming IDEs, data analysis tools for model training and validation, machine learning libraries, and UI design software for the scheduling dashboard. These tools are required throughout the full development lifecycle from data preprocessing to deployment.

7.3 Cloud Services

Cloud hosting is required for the backend API, ML inference service, and database storage. A managed cloud platform such as AWS or Google Cloud Platform provides scalable compute resources for real-time prediction and alert dispatch. Additional cloud storage is required for historical repair records, weather and traffic logs, and model artifacts.

7.4 Human Effort

The primary human effort categories include a backend developer for API integration and system implementation, an ML engineer for data preprocessing, model training, and evaluation, a UI/UX designer for the scheduling dashboard, and a QA engineer for end-to-end system testing and performance validation.

7.5 Overall Budget Summary

Table 7.1 presents the detailed proposed budget breakdown.

Category	Description	Estimated Cost
API Services	OpenWeatherMap and traffic API subscription keys for real-time data access	LKR 45,000
ML Libraries & Tools	Machine learning model development, training environment, and evaluation tools	LKR 25,000
Cloud Services	Cloud hosting for backend API, model inference service, and database storage	LKR 65,000
Software Development	IDE licenses, mapping SDK access, and data visualization software	LKR 40,000
Human Effort – Dev	Backend developer for API integration, ML engineer for model development	LKR 220,000
Human Effort – Design/QA	Dashboard UI/UX design, system testing, and quality assurance	LKR 100,000
Total Estimated Budget		**LKR 495,000**

The estimated total budget for developing the Environmental and Repair Time Analysis Module is approximately LKR 495,000. This budget covers essential API subscriptions, software tools, cloud resources, and human expertise required for module implementation. The investment is justified by the system's potential to significantly improve road maintenance scheduling efficiency, reduce repair rework, and minimize disruption to road users across Sri Lanka.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure outlines the complete task decomposition for this research module. Each task corresponds to a deliverable directly supporting the module objectives.

Task	Task Description	Timeline	Deliverable
T1	Conduct literature review on environmental analysis, traffic modeling, and repair scheduling systems	March 2026	Literature review chapter
T2	Collect stakeholder requirements – interviews with RDA officers and road maintenance engineers	March – April 2026	Stakeholder requirements report
T3	Integrate weather data API and implement real-time weather retrieval for defect locations	April 2026	Weather data integration module
T4	Integrate traffic flow API and develop traffic density analysis pipeline	April – May 2026	Traffic analysis module
T5	Collect and preprocess historical repair duration records for ML model training	May 2026	Annotated repair duration dataset
T6	Train and validate ML repair duration prediction model using weather, traffic, and surface features	May – June 2026	Trained prediction model
T7	Evaluate model accuracy and performance using MAPE, RMSE metrics on test dataset	June 2026	Model evaluation report
T8	Develop optimal start time recommendation engine integrating all environmental signals	July 2026	Scheduling engine module
T9	Implement alternate route suggestion module using mapping and directions APIs	July – August 2026	Route suggestion module

T10	Build scheduling dashboard for municipal authorities with maps, timelines, and alerts	August – September 2026	Web dashboard prototype
T11	Integrate module with defect detection and repair estimation components and conduct end-to-end testing	October 2026	Integration test report
T12	Prepare final documentation, evaluation results, and research paper	November 2026	Final research report

9. GANTT CHART

The following Gantt chart illustrates the project milestone timeline for the Environmental and Repair Time Analysis Module, aligned to the 2026 Regular Batch Research Project Plan.

Milestone / Deliverable	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Proposal Submission	●								
Literature Review Complete	●								
Weather & Traffic API Integration		●							
Dataset Collection Complete		●							
Progress Presentation 1			●						
ML Model Training Complete			●						
Model Evaluation Complete				●					
Scheduling Engine Implemented					●				
Route Suggestion Module Ready					●				
Progress Presentation 2						●			
Dashboard Prototype Ready							●		
System Integration Complete								●	

Final Report Submission									•
Final Presentation & Viva									•

Figure 9.1 – Project Gantt Chart Timeline (2026 Regular Batch Research Project Plan)

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11. APPENDICES

Appendix A – Survey Questionnaire for Road Maintenance Officers

Section A: Respondent Background

15. What is your role in road maintenance operations?

- Road Maintenance Engineer
- Planning Officer / Project Manager
- Field Supervisor
- Municipal Council Officer

16. How many years of experience do you have in road maintenance or civil infrastructure?

- Less than 2 years
- 2–5 years
- More than 5 years

Section B: Current Repair Scheduling Practices

17. How is repair scheduling currently determined in your operations?

- Based on field engineer judgment and experience
- Following a fixed maintenance schedule
- Using a digital planning or project management tool
- No formal scheduling process

18. What factors do you currently consider when scheduling a road repair?

- Weather conditions
- Traffic volume and peak hours
- Material and crew availability
- Urgency of defect severity

19. Have delays due to adverse weather or high traffic caused repair rework or extended closures?

- Yes, frequently
- Occasionally

- Rarely
- Never

Section C: Technology Adoption and Expectations

20. Would an AI-powered system that recommends optimal repair timing based on weather and traffic be useful?

- Yes
- No
- Not sure

21. What features would you expect from a smart repair scheduling system? (Select all that apply)

- Real-time weather conditions for repair sites
- Traffic density analysis and peak hour identification
- Automated repair duration estimates
- Optimal start time recommendations
- Alternate route suggestions for road users
- Automated stakeholder notification and alerts

Appendix B – Interview Questions for Road Engineers and RDA Officers

22. How do you currently estimate the time required to complete a road repair of a given size?
23. What environmental or external factors most significantly affect repair efficiency in your experience?
24. Have you ever had to abandon or repeat a repair due to weather or traffic issues? If so, how frequently?
25. How do you currently manage traffic disruption during road repair operations?
26. Are alternate routes provided to road users during road closures, and how are these communicated?
27. Would you be willing to use an automated scheduling tool that recommends start times based on weather and traffic data?
28. What concerns do you have about relying on an AI-based system for repair scheduling recommendations?

Appendix F – Risk Analysis and Mitigation Plan

ID	Risk Description	Impact	Mitigation Strategy
R1	Limited availability of historical road repair duration data from Sri Lankan authorities	**High**	Supplement with expert interviews, structured data collection from RDA, and transfer learning from international datasets
R2	Weather API rate limits or service unavailability during active scheduling sessions	**Medium**	Implement local caching of recent weather data and configure fallback to cached forecast data when live API is unavailable
R3	Traffic data inaccuracy in rural or low-connectivity regions of Sri Lanka	**High**	Combine API data with local sensor feeds and historical traffic patterns; flag lowconfidence estimates to users
R4	ML model underfitting due to high variability in road surface types and defect combinations	**High**	Apply ensemble learning techniques, include surface type as categorical feature, and retrain periodically with new operational data
R5	Road users ignoring alternate route suggestions due to low adoption of the notification system	**Medium**	Partner with navigation app providers (e.g., Waze, Google Maps) for integration; deploy multi-channel alerts (SMS, app, radio)
R6	Privacy concerns related to collection of location and traffic behavioral data	**Medium**	Anonymize all location records, comply with Sri Lankan data protection guidelines, and obtain informed consent from data contributors

Appendix G – Ethical Consent Form for Data Collection

Research Project: ROAD INSPECTOR – Automated Road Quality Inspection and Repair Estimation System

Researcher: IT22113122

Institution: Sri Lanka Institute of Information Technology

Purpose of Research: This study involves the collection of road repair duration records, environmental condition data, and traffic observations from road maintenance operations. The data will be used to train and validate machine learning models for repair duration prediction and scheduling optimization.

Voluntary Participation: Your participation in this study is entirely voluntary. You may withdraw from the study at any time without giving a reason.

Data Usage: All collected data will be anonymized and used strictly for academic research purposes. No personally identifiable information will be shared with third parties. Repair records will be stored securely and used only for model training and validation.

Confidentiality: Your identity and organization details will remain strictly confidential throughout the research process. All data will be stored in encrypted cloud storage accessible only to the research team.

Data Retention: All collected data will be retained for the duration of the research project and disposed of securely upon project completion in compliance with institutional guidelines.

Participant Name	Signature	Date

ROAD INSPECTOR

Automated Road Quality Inspection and Repair Estimation System

R26-IT-023

Project Proposal Report

MANATHUNGA M A D V G – IT22207968

B.Sc. (Hons) Degree in Information Technology
Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
MANATHUNGA M A D V G	IT22207968	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Supervisor Name	Date	Signature
SMB Harshanath		

ABSTRACT

Road maintenance planning in Sri Lanka and similar developing regions faces significant operational challenges arising from the absence of data-driven scheduling tools. Existing road inspection systems focus primarily on defection detection and surface damage assessment but rarely integrate environmental and contextual factors that directly affect repair efficiency. Weather conditions, traffic flow, and surface characteristics are critical determinants of repair duration, yet they are routinely excluded from maintenance planning processes, resulting in inefficient scheduling, extended road closures, and unnecessary disruption to road users.

This research proposes the Environmental and Repair Time Analysis Module, Component 4 of the ROAD INSPECTOR system. Upon receiving defect location data from the defect detection module, this component collects and analyzes real-time weather conditions, traffic flow patterns, and road surface characteristics to predict repair duration using a trained machine learning model. The system then recommends an optimal repair start time that aligns with favorable weather windows and minimum traffic periods, while also suggesting alternate routes for road users to minimize disruption during active repair operations.

The proposed module is implemented using weather and traffic APIs, machine learning regression techniques, and route planning services integrated within a cloud-hosted backend. The expected outcome is an intelligent scheduling support system that enables road maintenance authorities to plan repairs more accurately, reduce road user disruption, and respond proactively to environmental constraints. The system contributes to SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action) by promoting smarter and more responsive road infrastructure management.

Keywords: Road Maintenance Scheduling, Repair Time Prediction, Weather Analysis, Traffic Flow, Alternate Route Suggestion, Machine Learning, Smart Infrastructure

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Special thanks go to the officers and engineers of the Sri Lanka Road Development Authority (RDA) and municipal maintenance departments whose practical insights helped in understanding the real-world challenges related to road repair scheduling and environmental impact assessment.

I would also like to acknowledge the collaborative nature of the ROAD INSPECTOR project team, whose collective contributions to the overall system design and architecture greatly informed the scope and direction of this module.

Finally, I would like to express my heartfelt gratitude to my family and friends for their constant encouragement, patience, and moral support throughout the research process. This work would not have been possible without the support and motivation provided by all of them.

TABLE OF CONTENTS

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENTS 5
1. INTRODUCTION 7
1.1 Background and Context 7
1.2 Literature Review 7
1.3 Research Gap 9
2. OBJECTIVES 10
2.1 Main Objective 10
2.2 Specific Objectives 10
3. METHODOLOGY 12
3.1 Research Approach 12
3.2 Development Stages 12
3.3 Technologies and Tools 13
3.4 Data Collection Methods 14
4. HIGH-LEVEL SYSTEM ARCHITECTURE 15
4.1 Architecture Overview 15
Data Ingestion Layer 16
Analysis and Prediction Layer 17
Scheduling and Routing Layer 17
Notification and Presentation Layer 17
4.2 System Component Descriptions 18
4.3 Data Flow 18
4.4 Scalability and Security 19
5. USER REQUIREMENTS 20
5.1 Stakeholder Identification 20
5.2 Functional Requirements 20
5.3 Non-Functional Requirements 21
6. COMMERCIALIZATION PLAN 22
6.1 Target Market and Customer Persona 22
6.2 Value Proposition 22

6.3 Revenue Model 23
6.4 Cost Estimation 23
6.5 Pricing Strategy 23
6.6 Competitive Advantage 23
6.7 Intellectual Property Considerations 24
7. BUDGET AND JUSTIFICATION 25
7.1 API Services 25
7.2 Software Licenses 25
7.3 Cloud Services 25
7.4 Human Effort 25
7.5 Overall Budget Summary 26
8. WORK BREAKDOWN STRUCTURE (WBS) 27
9. GANTT CHART 28
10. REFERENCES LIST 29
11. APPENDICES 30
Appendix A – Survey Questionnaire for Road Maintenance Officers 30
Section A: Respondent Background 30
Section B: Current Repair Scheduling Practices 30
Section C: Technology Adoption and Expectations 31
Appendix B – Interview Questions for Road Engineers and RDA Officers 31
Appendix F – Risk Analysis and Mitigation Plan 32
Appendix G – Ethical Consent Form for Data Collection 32

1. INTRODUCTION

1.1 Background and Context

Road infrastructure is a critical component of Sri Lanka's transportation network, supporting daily commutes, tourism, and commercial logistics. The maintenance of road surfaces is an ongoing requirement, with defects such as potholes, cracks, and surface erosion frequently occurring due to heavy rainfall, high traffic loads, and aging road materials. Timely and efficient repair of these defects is essential to ensure road safety and minimize economic losses.

Despite the importance of efficient road maintenance, scheduling decisions in Sri Lanka are predominantly made based on manual assessments and the subjective judgment of field engineers. Environmental factors such as rainfall, temperature, and humidity are rarely systematically factored into repair planning. Similarly, traffic flow conditions at repair sites are not routinely analyzed to determine the least disruptive scheduling window. The absence of these analytical inputs leads to frequent delays, extended road closures, and higher-than-necessary material wastage.

The ROAD INSPECTOR system addresses these challenges through a modular architecture that combines road defect detection, repair area estimation, material calculation, and environmental analysis into a unified platform. This proposal focuses on Component 4, the Environmental and Repair Time Analysis Module, which is responsible for analyzing weather conditions, traffic flow, and surface characteristics to estimate repair duration, recommend optimal scheduling times, and suggest alternate routes for affected road users.

This research aligns with SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action) by promoting data-driven, environmentally aware road maintenance practices that reduce waste, minimize disruption, and support sustainable infrastructure management.

1.2 Literature Review

A growing body of research has explored the application of computer vision, deep learning, and sensor-based technologies for road condition monitoring and defect detection. However, the integration of environmental and traffic context into repair scheduling decisions remains an underexplored area.

Harikrishnan and Gopi (2017) developed a vehicle-based road condition monitoring system using smartphone sensors and GPS, demonstrating the feasibility of automated data collection for pavement assessment. However, the study did not address repair duration prediction or scheduling optimization.

Maeda et al. (2018) proposed a deep learning framework using VGG16 for road damage detection from smartphone images, achieving strong classification performance across multiple defect types. While the detection accuracy was notable, the system did not incorporate environmental factors affecting repair operations or provide actionable maintenance scheduling support.

Fan et al. (2019) investigated the use of unmanned aerial vehicles (UAVs) for pavement distress surveying, enabling efficient large-area road surface assessment. Although the approach improved spatial coverage and data quality, the system remained focused on inspection rather than repair planning or environmental integration.

Arya et al. (2021) demonstrated the effectiveness of YOLO-based object detection for real-time pothole and crack classification from road imagery. The model achieved competitive accuracy metrics but did not provide downstream repair time estimation or scheduling recommendations.

Chatterjee and Ghosh (2023) proposed an IoT-based road condition monitoring framework incorporating environmental sensors to assess surface deterioration rates. While the system collected temperature and moisture data, it did not apply predictive modeling to generate repair duration estimates or suggest alternate routes for road users.

Table 1.1 summarizes the key characteristics and limitations of existing systems in comparison with the proposed module.

Study / System	Focus Area	Key Technology	Limitation
Harikrishnan et al. (2022)	Pothole detection	CNN, Smartphone camera	No repair estimation
Maeda et al. (2018)	Road damage detection	Deep Learning, VGG16	No scheduling or alerting
Fan et al. (2019)	Pavement distress survey	UAV, image processing	No traffic/weather context
Arya et al. (2021)	Crack classification	YOLO, ResNet	No material or time estimation
Chatterjee et al. (2023)	Road condition monitoring	IoT sensors, ML	No alternate route suggestion
Proposed Component	Env. & repair time analysis	ML, weather API, traffic data	Addresses all identified gaps

1.3 Research Gap

Although significant progress has been made in road defect detection and surface condition monitoring, several critical gaps remain unaddressed in existing literature and commercial systems.

The most significant gap is the absence of integrated environmental analysis in repair scheduling systems. Existing tools detect road defects and estimate surface damage but do not analyze weather conditions or traffic flow to determine when repairs should be carried out. As a result, maintenance teams frequently schedule repairs during adverse weather or peak traffic periods, reducing effectiveness and increasing public inconvenience.

A related gap is the lack of machine learning models trained specifically for repair duration prediction incorporating environmental variables. While construction project duration prediction has been studied in other domains, road repair duration estimation using weather, traffic, and surface data has not been systematically addressed in the context of developing countries such as Sri Lanka.

Additionally, no existing system provides integrated alternate route recommendations for road users during active repair periods. This function is essential for minimizing disruption, particularly on roads with limited alternative access routes, but it has not been included within road inspection platforms.

Table 1.2 presents a detailed comparison of existing systems against the proposed module across critical functional dimensions.

Feature	Harikrishnan	Maeda	Fan	Arya	Proposed
Weather Impact Analysis	No	No	No	No	**Yes**
Traffic Flow Integration	No	No	No	No	**Yes**
Repair Duration Prediction	No	No	No	No	**Yes**
Alternate Route Suggestion	No	No	No	No	**Yes**
Optimal Start Time Recommendation	No	No	No	No	**Yes**
Real-Time Stakeholder Alerts	Partial	No	No	No	**Yes**

2. OBJECTIVES

2.1 Main Objective

To design and develop an Environmental and Repair Time Analysis Module that, upon receiving defect location data, analyzes real-time weather conditions, traffic flow patterns, and road surface characteristics using machine learning techniques to predict repair duration, recommend an optimal repair start time, and suggest alternate routes for road users, within the context of the ROAD INSPECTOR automated road quality inspection and repair estimation system.

The objective is:

- Specific – weather-aware, traffic-sensitive repair duration prediction and scheduling for Sri Lankan road maintenance operations
- Measurable – prediction error rate (MAPE), alert notification latency, and alternate route suggestion accuracy
- Achievable – based on well-established machine learning regression, REST API integration, and route planning techniques
- Relevant – directly supports efficient road maintenance scheduling and minimization of road user disruption
- Time-bound – within the defined research project duration from March to November 2026

2.2 Specific Objectives

1. To collect and integrate real-time weather data including rainfall probability, temperature, humidity, and wind conditions for the specific defect location received from the road defect detection module.
2. To analyze traffic density and flow patterns at and around the repair site using traffic data APIs, identifying peak and off-peak traffic periods to support scheduling decisions.
3. To develop and train a machine learning model for repair duration prediction that incorporates weather conditions, traffic flow, defect size and depth, and road surface characteristics as input features.
4. To design an optimal scheduling engine that combines weather forecasts, traffic predictions, and duration estimates to recommend the most suitable repair start time with minimal environmental and traffic risk.
5. To implement an alternate route suggestion module that identifies and presents viable detour routes for road users and vehicles affected by the repair site closure.
6. To develop an automated alert and notification system that communicates repair schedules, estimated durations, closure areas, and alternate routes to relevant stakeholders in real time.

Table 2.1 summarizes the specific objectives and their expected outcomes.

Specific Objective	Expected Outcome
Collect and integrate real-time weather data for defect locations	Accurate weather-aware repair difficulty assessment
Analyze traffic density and flow patterns at repair sites	Traffic-optimized repair scheduling to minimize disruption
Predict repair duration using ML models incorporating environmental factors	Data-driven time estimates replacing manual guesswork
Recommend optimal repair start time based on conditions	Maximally efficient scheduling with minimal traffic impact
Suggest alternate routes for road users during repair	Reduced road user disruption and improved safety
Generate automated alerts and repair scheduling reports	Timely stakeholder notifications and operational transparency

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) methodology, which focuses on designing, developing, and evaluating innovative technological artifacts to address real-world problems. In this study, the artifact is the Environmental and Repair Time Analysis Module of the ROAD INSPECTOR system. The research process encompasses five structured stages: problem identification, requirements analysis, system design, module development and integration, and performance evaluation.

3.2 Development Stages

Stage 1 – Requirements Analysis: Requirements will be identified through literature reviews, structured interviews with road maintenance engineers and officers from the Sri Lanka Road Development Authority (RDA), and analysis of existing maintenance scheduling workflows. The focus will be on understanding current environmental assessment practices, repair duration estimation methods, and traffic impact management approaches.

Stage 2 – System Architecture Design: The module architecture will be designed to integrate weather data APIs, traffic flow APIs, a machine learning prediction engine, a scheduling optimization component, and an alternate route recommendation service. Both component-level and data flow architectures will be documented.

Stage 3 – Data Collection and Preprocessing: Historical road repair records will be collected from the RDA and municipal authorities to build the repair duration training dataset. Weather and traffic data corresponding to each historical repair will be retrieved via APIs to construct feature sets. Data will be cleaned, normalized, and annotated for model training.

Stage 4 – Machine Learning Model Development: A supervised regression model will be trained to predict repair duration in hours given inputs of defect size and depth, road surface type, weather conditions, traffic density, and crew size. Multiple algorithms including Random Forest, Gradient Boosting (XGBoost), and Linear Regression will be evaluated, and the best-performing model will be selected based on Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE).

Stage 5 – System Integration and Evaluation: All sub-components including weather analysis, traffic analysis, ML prediction, scheduling engine, and alternate route suggestion will be integrated within a unified cloud-hosted platform. The module will receive defect location data from Component 1 (defect detection) and return scheduling recommendations to the ROAD INSPECTOR dashboard. End-to-end performance evaluation will be conducted using simulated and real repair scenarios.

3.3 Technologies and Tools

Table 3.1 outlines the selected technologies and tools for module development.

Component	Technology / Tool	Purpose
Weather Data Integration	OpenWeatherMap API / WeatherAPI	Retrieve real-time and forecast weather conditions
Traffic Flow Analysis	Google Maps Platform / HERE Traffic API	Obtain live traffic density and speed data
ML Repair Time Prediction	Python, Scikit-learn, XGBoost	Train and deploy repair duration prediction model
Surface Characteristic Analysis	OpenCV, image classification models	Analyze road surface type and damage severity
Alternate Route Engine	Google Maps Directions API / OSRM	Compute and display alternate routes for road users
Backend API	Node.js, Express.js	REST API for component communication
Database	PostgreSQL / MongoDB	Store environmental logs, predictions, and route data
Dashboard / Notification	React.js, Firebase Cloud Messaging	Display scheduling and route info; send alerts
Cloud Hosting	AWS / Google Cloud Platform	Scalable model deployment and data storage

3.4 Data Collection Methods

Table 3.2 outlines the data collection methods and sources used in this research.

Data Type	Collection Method	Source
Weather conditions	API integration and historical data retrieval	OpenWeatherMap, Sri Lanka Meteorological Dept.
Traffic flow patterns	Live traffic API calls and sensor logs	Google Maps Traffic API, HERE API
Road surface characteristics	Image classification from defect detection module	Component 1 output (defect detection module)
Repair duration records	Structured interviews and maintenance logs	Sri Lanka Road Development Authority (RDA)
Historical repair data	Database records from public authorities	RDA maintenance records and project reports
Alternate route data	Route computation via mapping APIs	Google Maps Directions API, OpenStreetMap

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Architecture Overview

The Environmental and Repair Time Analysis Module adopts a layered architecture comprising four functional layers: the Data Ingestion Layer, the Analysis and Prediction Layer, the Scheduling and Routing Layer, and the Notification and Presentation Layer. This layered design ensures clear separation of concerns, scalability, and modular integration with other ROAD INSPECTOR components.

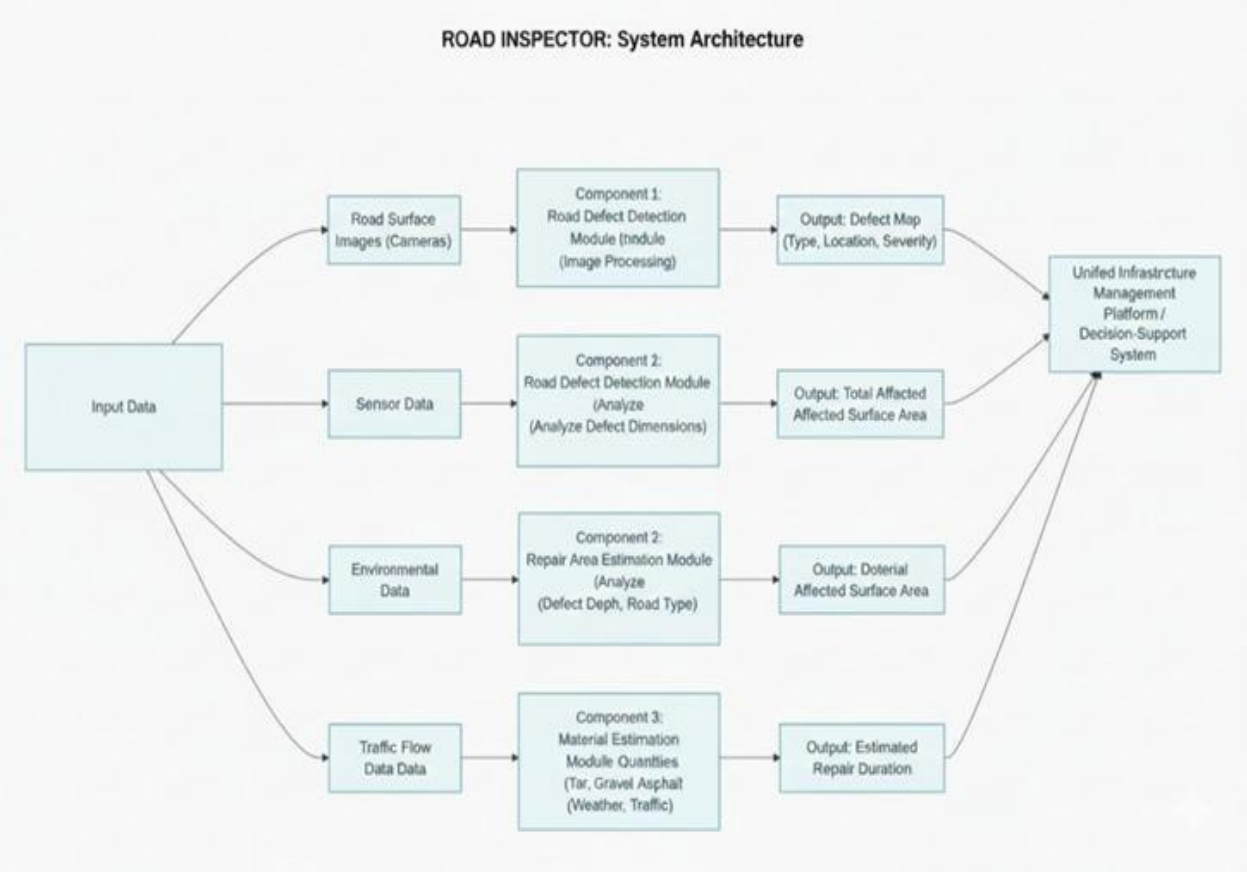




Figure 4.1 System Architecture Diagram

Data Ingestion Layer

- Defect location and surface metadata received from Component 1 (Road Defect Detection Module)
- Real-time weather conditions retrieved via OpenWeatherMap or WeatherAPI
- Live traffic density and flow data retrieved via Google Maps Traffic API or HERE Traffic API
- Historical repair duration records from RDA maintenance databases

Analysis and Prediction Layer

- Weather condition classification and repair feasibility assessment
- Traffic density scoring and peak period identification
- Surface characteristic processing from defect detection module output
- ML repair duration prediction model inference

Scheduling and Routing Layer

- Optimal repair start time recommendation engine
- Alternate route computation via Directions API or OSRM
- Repair schedule generation with weather and traffic overlays

Notification and Presentation Layer

- Automated push notifications to road users, maintenance teams, and municipal authorities
- Scheduling and route map dashboard for road maintenance planners
- Integration with ROAD INSPECTOR central operations panel

4.2 System Component Descriptions

Table 4.1 provides detailed descriptions of all key module components.

Component	Description
Weather Analysis Engine	Retrieves real-time and forecast weather data for the defect location via API. Analyzes rainfall probability, temperature, humidity, and wind conditions to assess repair feasibility and difficulty level.
Traffic Flow Analyzer	Collects live traffic density, speed, and congestion data for roads surrounding the repair site. Identifies peak and off-peak periods to recommend the least disruptive repair window.
Surface Characteristic Processor	Receives defect metadata and surface type information from the Road Defect Detection Module. Classifies surface material (asphalt, concrete, gravel) and evaluates road condition severity as input to duration prediction.
ML Repair Duration Predictor	Applies a trained machine learning model to estimate repair time based on defect size, depth, surface type, weather conditions, traffic flow, and crew capacity. Outputs estimated start-to-completion time in hours.
Optimal Scheduling Engine	Combines weather forecasts, traffic predictions, and duration estimates to recommend the best repair start time. Prioritizes low-traffic periods with favorable weather conditions.
Alternate Route Suggestion Module	Uses mapping APIs to identify and present alternate road routes for vehicles and pedestrians affected by the repair closure. Routes are displayed on a map interface and sent via notifications.
Alert and Notification System	Pushes real-time alerts to road users, municipal authorities, and maintenance teams. Notifications include repair start time, estimated duration, closure area, and recommended detour routes.
Scheduling Dashboard	A web-based interface displaying repair schedules, weather and traffic summaries, alternate route maps, and duration predictions for municipal road maintenance teams.

4.3 Data Flow

7. The Road Defect Detection Module (Component 1) transmits defect location coordinates, surface type, defect dimensions, and severity classification to the Environmental and Repair Time Analysis Module via REST API.
8. The Weather Analysis Engine retrieves current and 48-hour forecast weather data for the defect coordinates and generates a weather condition score and repair feasibility rating.
9. The Traffic Flow Analyzer collects live traffic density and speed data for the roads surrounding the defect location and identifies peak and off-peak traffic windows over the next 72 hours.

10. The Surface Characteristic Processor ingests defect metadata and classifies the road surface material and damage level to parameterize the duration prediction model.
11. The ML Repair Duration Predictor combines weather scores, traffic density, defect dimensions, and surface data to generate an estimated repair duration in hours.
12. The Scheduling Engine evaluates available weather and traffic windows over the planning horizon and recommends an optimal repair start time with the lowest combined weather and traffic risk.
13. The Alternate Route Suggestion Module calls the routing API to compute viable detour routes and formats them for display and notification.
14. The Notification System dispatches alerts to all relevant stakeholders containing repair schedule, estimated duration, closure boundaries, and recommended alternate routes.

4.4 Scalability and Security

The module is designed for cloud-based scalability, enabling simultaneous analysis for multiple repair sites across different geographic locations. Modular microservice architecture allows individual components such as weather retrieval, traffic analysis, and ML inference to be scaled independently based on demand. Horizontal scaling on the cloud backend ensures consistent performance during high-demand periods following major weather events or natural disasters that increase the volume of road defect reports.

Security measures include encrypted transmission of all API communications using TLS/HTTPS, role-based access control for the scheduling dashboard, anonymization of location data in compliance with Sri Lankan data protection guidelines, and secure cloud storage for all historical repair records and prediction logs.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The Environmental and Repair Time Analysis Module serves the following primary stakeholders.

- Road Maintenance Planners: Primary users of the scheduling dashboard who require accurate repair duration estimates, optimal start time recommendations, and weather and traffic overlays to plan repair operations efficiently.
- Municipal Road Authorities: Supervisory stakeholders who require visibility into repair schedules, expected completion times, and site closure durations for operational oversight and public communication.
- Vehicle Drivers and Road Users: Beneficiaries of alternate route suggestions and repair schedule notifications who need timely information to plan journeys and avoid closure areas.
- Maintenance Field Teams: Teams who receive work order information including scheduled start time, expected duration, and weather advisories to prepare equipment and manpower accordingly.
- ROAD INSPECTOR System Administrator: Manages module configuration, monitors API performance, and oversees ML model retraining cycles.

5.2 Functional Requirements

Table 5.1 details the functional requirements of the proposed module.

ID	Functional Requirement	Stakeholder
FR1	The system shall retrieve real-time weather conditions (temperature, rainfall, humidity) for each received defect location.	Maintenance Teams, Planners
FR2	The system shall analyze traffic density and movement patterns at and near the repair site.	Road Users, Planners
FR3	The system shall predict the estimated repair duration using a trained ML model incorporating weather, traffic, surface, and defect data.	Maintenance Teams, Authorities
FR4	The system shall recommend an optimal repair start time that minimizes traffic disruption and aligns with favorable weather windows.	Maintenance Planners
FR5	The system shall compute and display alternate road routes for vehicles and pedestrians affected by the repair closure.	Road Users, Public

FR6	The system shall generate automated alerts and notifications to stakeholders containing repair schedule, estimated duration, and detour information.	All Stakeholders
FR7	The system shall display a scheduling dashboard showing repair timelines, weather summaries, traffic overlays, and alternate routes.	Municipal Authorities
FR8	The system shall integrate with the defect detection and repair estimation modules to receive defect location and surface data.	System Components

5.3 Non-Functional Requirements

Table 5.2 details the non-functional requirements of the proposed module.

Category	Requirement	Description
Performance	**Weather data refresh ≤ 10 min**	Weather conditions must be refreshed at intervals no greater than ten minutes during active scheduling sessions.
Performance	**Prediction response ≤ 5 seconds**	Repair duration predictions and scheduling recommendations must be returned within five seconds of receiving defect input.
Accuracy	**Duration prediction error ≤ 15%**	The ML model must achieve a mean absolute percentage error of no more than 15% on the test dataset for repair duration estimates.
Usability	**Non-technical dashboard access**	The scheduling dashboard must be operable by municipal staff without technical expertise, with clearly labeled controls and visual maps.
Reliability	**Offline data caching**	The system must cache the most recent weather and traffic data locally to maintain scheduling recommendations during brief API unavailability.
Scalability	**Multi-site concurrent support**	The architecture must support simultaneous scheduling analysis for multiple repair sites across different geographic locations.
Security	**Encrypted data transmission**	All API communications and stored scheduling data must use TLS/HTTPS encryption with role-based access control.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the Environmental and Repair Time Analysis Module includes government road maintenance authorities, municipal engineering departments, national highway operators, and private road infrastructure contractors who require data-driven scheduling and disruption management tools.

A typical customer persona would be a road maintenance planning officer in a municipal council or the Sri Lanka Road Development Authority who needs accurate repair duration estimates and scheduling recommendations to coordinate field teams, manage road closures, and communicate repair timelines to the public.

Table 6.1 summarizes the target market segmentation.

Segment	Target Customer	Use Case
Primary	Municipal road maintenance authorities	Repair scheduling, duration planning, and resource allocation
Secondary	National highway and expressway operators	Large-scale road maintenance coordination and disruption management
Tertiary	Private infrastructure contractors	Project scheduling support and weather-risk planning
Emerging	Navigation and mapping application providers	Integration of road closure and alternate route data into user applications

6.2 Value Proposition

The proposed module delivers measurable value across multiple dimensions of road maintenance operations.

- Data-driven repair scheduling that replaces subjective manual planning with accurate, ML-powered duration estimates
- Weather-aware scheduling that avoids repair operations during adverse conditions, reducing rework and material wastage
- Traffic-optimized timing that minimizes road closure impact by scheduling repairs during off-peak periods
- Automated alternate route suggestions that reduce road user disruption and improve public communication

- Real-time stakeholder alerts that provide proactive information to drivers, maintenance teams, and municipal authorities

6.3 Revenue Model

The module is offered as part of the broader ROAD INSPECTOR Software-as-a-Service (SaaS) platform. Specific revenue streams include subscription fees for municipal and government road authorities based on the number of active repair sites monitored, integration licensing fees for navigation application providers who incorporate alternate route data, and premium scheduling analytics packages offering historical repair performance reporting and model performance dashboards.

6.4 Cost Estimation

Table 6.2 summarizes the estimated development and operational costs for this module.

Cost Category	Development Cost	Annual Operating Cost
Weather & Traffic API subscriptions	LKR 46,650	LKR 93,300
Software licenses and development tools	LKR 37,320	LKR 24,880
Cloud hosting and ML model deployment	LKR 62,200	LKR 93,300
Human effort (development and engineering)	LKR 310,000	LKR 186,600
Total Estimated Cost	**LKR 456,170**	**LKR 398,080**

6.5 Pricing Strategy

A tiered pricing model is proposed.

- Basic Plan – Small municipal councils with up to ten concurrent repair sites: weather analysis, duration prediction, and notification features
- Standard Plan – Medium authorities with up to thirty sites, including full scheduling engine and alternate route suggestions
- Enterprise Plan – National highway operators and large government departments with unlimited sites, advanced ML reporting, custom alert integrations, and API access for third-party applications

6.6 Competitive Advantage

- First integrated system combining weather analysis, traffic flow, ML duration prediction, and route suggestion within a road maintenance scheduling context
- ML model trained specifically on Sri Lankan road repair data incorporating local surface materials, climate patterns, and traffic behavior
- Proactive scheduling recommendations that prevent repair inefficiency rather than reporting problems after they occur

- Seamless integration with the broader ROAD INSPECTOR platform providing end-to-end road inspection and maintenance decision support

6.7 Intellectual Property Considerations

The proposed module may generate intellectual property assets including the trained ML model for repair duration prediction incorporating Sri Lankan road and environmental data, the scheduling optimization algorithm combining weather and traffic signals, and the integrated alternate route suggestion workflow. The development team may consider software copyright protection, proprietary algorithm documentation, and potential patent registration for the unified environmental-aware scheduling methodology.

7. BUDGET AND JUSTIFICATION

The following budget outlines the estimated costs required to design, develop, and implement the Environmental and Repair Time Analysis Module. The budget covers API subscriptions, software tools, cloud infrastructure, and human effort required for module development and deployment.

7.1 API Services

Weather and traffic API access keys are essential for retrieving real-time and historical environmental and traffic data. OpenWeatherMap provides current conditions, forecasts, and historical weather records. A traffic API such as Google Maps Platform or HERE provides live congestion and speed data. These subscriptions are required both during development for testing and in production for live operation.

7.2 Software Licenses

Development environment licenses include programming IDEs, data analysis tools for model training and validation, machine learning libraries, and UI design software for the scheduling dashboard. These tools are required throughout the full development lifecycle from data preprocessing to deployment.

7.3 Cloud Services

Cloud hosting is required for the backend API, ML inference service, and database storage. A managed cloud platform such as AWS or Google Cloud Platform provides scalable compute resources for real-time prediction and alert dispatch. Additional cloud storage is required for historical repair records, weather and traffic logs, and model artifacts.

7.4 Human Effort

The primary human effort categories include a backend developer for API integration and system implementation, an ML engineer for data preprocessing, model training, and evaluation, a UI/UX designer for the scheduling dashboard, and a QA engineer for end-to-end system testing and performance validation.

7.5 Overall Budget Summary

Table 7.1 presents the detailed proposed budget breakdown.

Category	Description	Estimated Cost
API Services	OpenWeatherMap and traffic API subscription keys for real-time data access	LKR 45,000
ML Libraries & Tools	Machine learning model development, training environment, and evaluation tools	LKR 25,000
Cloud Services	Cloud hosting for backend API, model inference service, and database storage	LKR 65,000
Software Development	IDE licenses, mapping SDK access, and data visualization software	LKR 40,000
Human Effort – Dev	Backend developer for API integration, ML engineer for model development	LKR 220,000
Human Effort – Design/QA	Dashboard UI/UX design, system testing, and quality assurance	LKR 100,000
Total Estimated Budget		**LKR 495,000**

The estimated total budget for developing the Environmental and Repair Time Analysis Module is approximately LKR 495,000. This budget covers essential API subscriptions, software tools, cloud resources, and human expertise required for module implementation. The investment is justified by the system's potential to significantly improve road maintenance scheduling efficiency, reduce repair rework, and minimize disruption to road users across Sri Lanka.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure outlines the complete task decomposition for this research module. Each task corresponds to a deliverable directly supporting the module objectives.

Task	Task Description	Timeline	Deliverable
T1	Conduct literature review on environmental analysis, traffic modeling, and repair scheduling systems	March 2026	Literature review chapter
T2	Collect stakeholder requirements – interviews with RDA officers and road maintenance engineers	March – April 2026	Stakeholder requirements report
T3	Integrate weather data API and implement real-time weather retrieval for defect locations	April 2026	Weather data integration module
T4	Integrate traffic flow API and develop traffic density analysis pipeline	April – May 2026	Traffic analysis module
T5	Collect and preprocess historical repair duration records for ML model training	May 2026	Annotated repair duration dataset
T6	Train and validate ML repair duration prediction model using weather, traffic, and surface features	May – June 2026	Trained prediction model
T7	Evaluate model accuracy and performance using MAPE, RMSE metrics on test dataset	June 2026	Model evaluation report
T8	Develop optimal start time recommendation engine integrating all environmental signals	July 2026	Scheduling engine module
T9	Implement alternate route suggestion module using mapping and directions APIs	July – August 2026	Route suggestion module
T10	Build scheduling dashboard for municipal authorities with maps, timelines, and alerts	August – September 2026	Web dashboard prototype
T11	Integrate module with defect detection and repair estimation components and conduct end-to-end testing	October 2026	Integration test report
T12	Prepare final documentation, evaluation results, and research paper	November 2026	Final research report

9. GANTT CHART

The following Gantt chart illustrates the project milestone timeline for the Environmental and Repair Time Analysis Module, aligned to the 2026 Regular Batch Research Project Plan.

Milestone / Deliverable	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Proposal Submission	●								
Literature Review Complete	●								
Weather & Traffic API Integration		●							
Dataset Collection Complete		●							
Progress Presentation 1			●						
ML Model Training Complete			●						
Model Evaluation Complete				●					
Scheduling Engine Implemented					●				
Route Suggestion Module Ready					●				
Progress Presentation 2						●			
Dashboard Prototype Ready							●		
System Integration Complete								●	
Final Report Submission									●
Final Presentation & Viva									●

Figure 9.1 – Project Gantt Chart Timeline (2026 Regular Batch Research Project Plan)

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11. APPENDICES

Appendix A – Survey Questionnaire for Road Maintenance Officers

Section A: Respondent Background

15. What is your role in road maintenance operations?

- Road Maintenance Engineer
- Planning Officer / Project Manager
- Field Supervisor
- Municipal Council Officer

16. How many years of experience do you have in road maintenance or civil infrastructure?

- Less than 2 years
- 2–5 years
- More than 5 years

Section B: Current Repair Scheduling Practices

17. How is repair scheduling currently determined in your operations?

- Based on field engineer judgment and experience
- Following a fixed maintenance schedule
- Using a digital planning or project management tool
- No formal scheduling process

18. What factors do you currently consider when scheduling a road repair?

- Weather conditions
- Traffic volume and peak hours
- Material and crew availability
- Urgency of defect severity

19. Have delays due to adverse weather or high traffic caused repair rework or extended closures?

- Yes, frequently
- Occasionally
- Rarely
- Never

Section C: Technology Adoption and Expectations

20. Would an AI-powered system that recommends optimal repair timing based on weather and traffic be useful?

- Yes
- No
- Not sure

21. What features would you expect from a smart repair scheduling system? (Select all that apply)

- Real-time weather conditions for repair sites
- Traffic density analysis and peak hour identification
- Automated repair duration estimates
- Optimal start time recommendations
- Alternate route suggestions for road users
- Automated stakeholder notification and alerts

Appendix B – Interview Questions for Road Engineers and RDA Officers

22. How do you currently estimate the time required to complete a road repair of a given size?
23. What environmental or external factors most significantly affect repair efficiency in your experience?
24. Have you ever had to abandon or repeat a repair due to weather or traffic issues? If so, how frequently?
25. How do you currently manage traffic disruption during road repair operations?
26. Are alternate routes provided to road users during road closures, and how are these communicated?
27. Would you be willing to use an automated scheduling tool that recommends start times based on weather and traffic data?
28. What concerns do you have about relying on an AI-based system for repair scheduling recommendations?

Appendix F – Risk Analysis and Mitigation Plan

ID	Risk Description	Impact	Mitigation Strategy
R1	Limited availability of historical road repair duration data from Sri Lankan authorities	**High**	Supplement with expert interviews, structured data collection from RDA, and transfer learning from international datasets
R2	Weather API rate limits or service unavailability during active scheduling sessions	**Medium**	Implement local caching of recent weather data and configure fallback to cached forecast data when live API is unavailable
R3	Traffic data inaccuracy in rural or low-connectivity regions of Sri Lanka	**High**	Combine API data with local sensor feeds and historical traffic patterns; flag low-confidence estimates to users
R4	ML model underfitting due to high variability in road surface types and defect combinations	**High**	Apply ensemble learning techniques, include surface type as categorical feature, and retrain periodically with new operational data
R5	Road users ignoring alternate route suggestions due to low adoption of the notification system	**Medium**	Partner with navigation app providers (e.g., Waze, Google Maps) for integration; deploy multi-channel alerts (SMS, app, radio)
R6	Privacy concerns related to collection of location and traffic behavioral data	**Medium**	Anonymize all location records, comply with Sri Lankan data protection guidelines, and obtain informed consent from data contributors

Appendix G – Ethical Consent Form for Data Collection

Research Project: ROAD INSPECTOR – Automated Road Quality Inspection and Repair Estimation System

Researcher: IT22207968

Institution: Sri Lanka Institute of Information Technology

Purpose of Research: This study involves the collection of road repair duration records, environmental condition data, and traffic observations from road maintenance operations. The data will be used to train and validate machine learning models for repair duration prediction and scheduling optimization.

Voluntary Participation: Your participation in this study is entirely voluntary. You may withdraw from the study at any time without giving a reason.

Data Usage: All collected data will be anonymized and used strictly for academic research purposes. No personally identifiable information will be shared with third parties. Repair records will be stored securely and used only for model training and validation.

Confidentiality: Your identity and organization details will remain strictly confidential throughout the research process. All data will be stored in encrypted cloud storage accessible only to the research team.

Data Retention: All collected data will be retained for the duration of the research project and disposed of securely upon project completion in compliance with institutional guidelines.

Participant Name	Signature	Date